

Jeric Cantos | Cat Gipetto

TravelTide Customer Segmentation

Version 1

Agenda

- Introduction
- Methodology
- Personas / Segments
- Next Steps

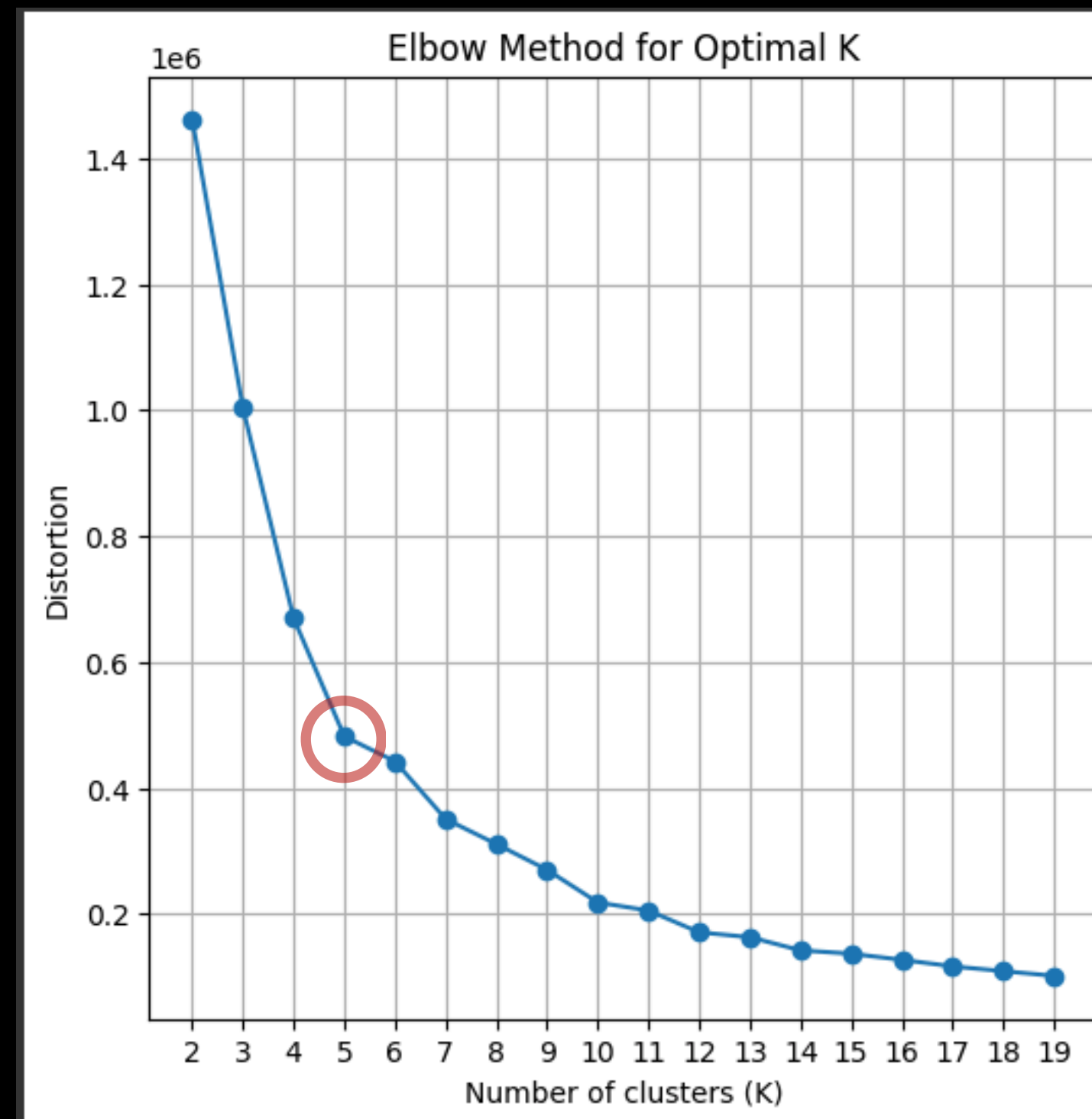


Methodology

ML Clustering

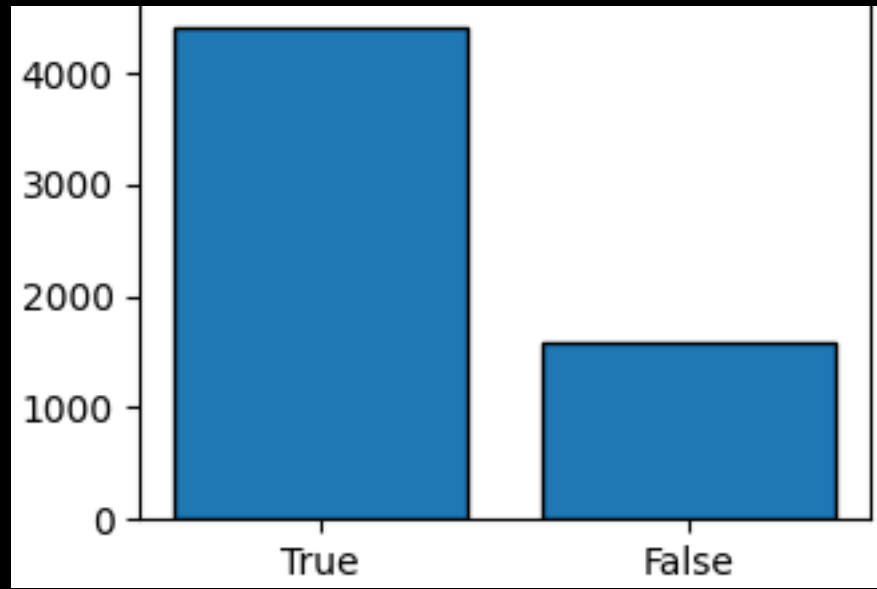
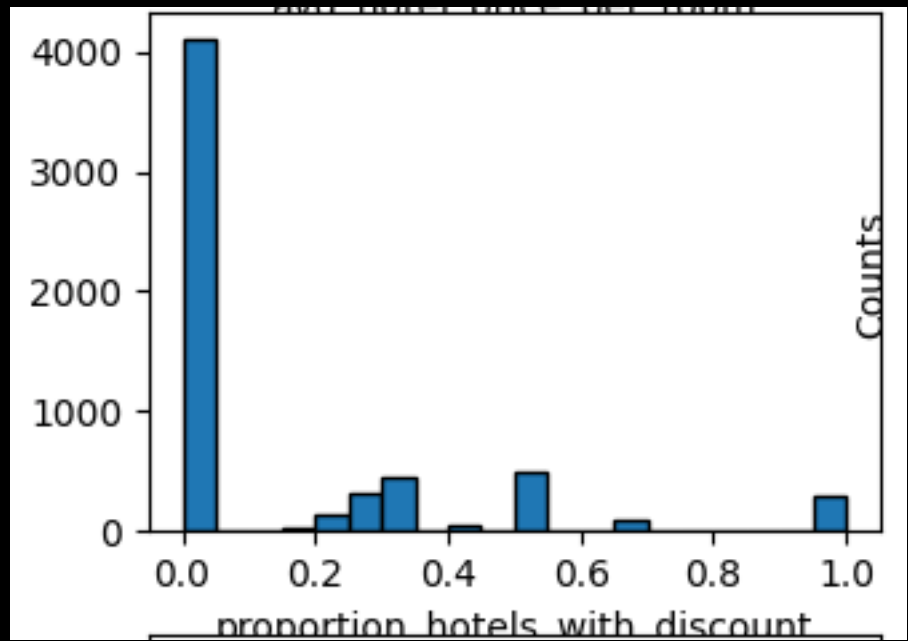
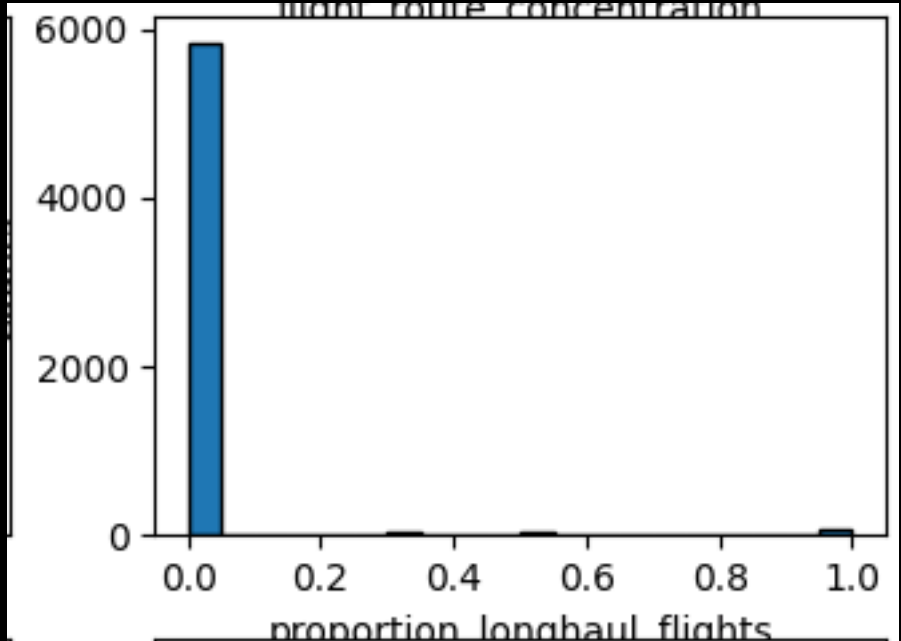
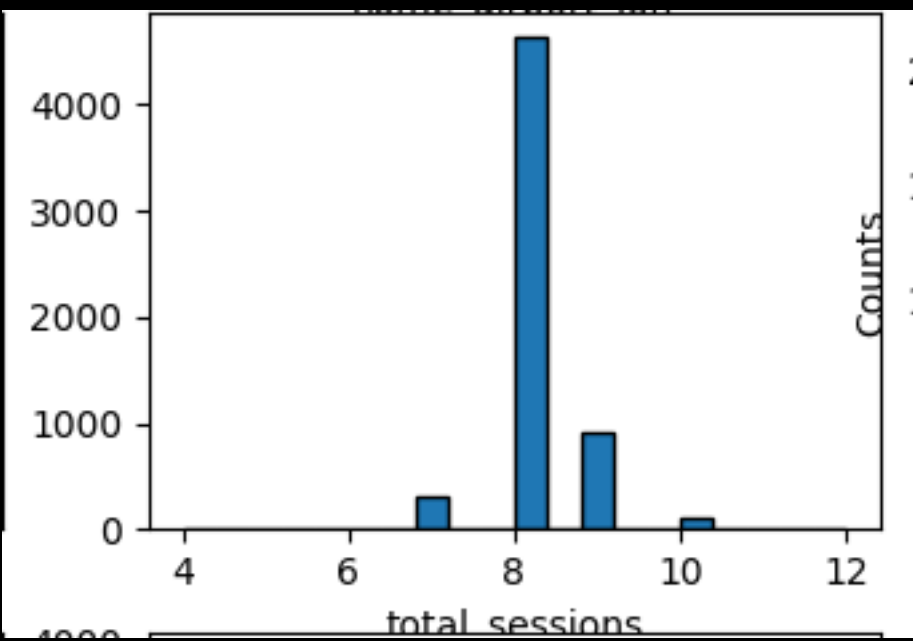
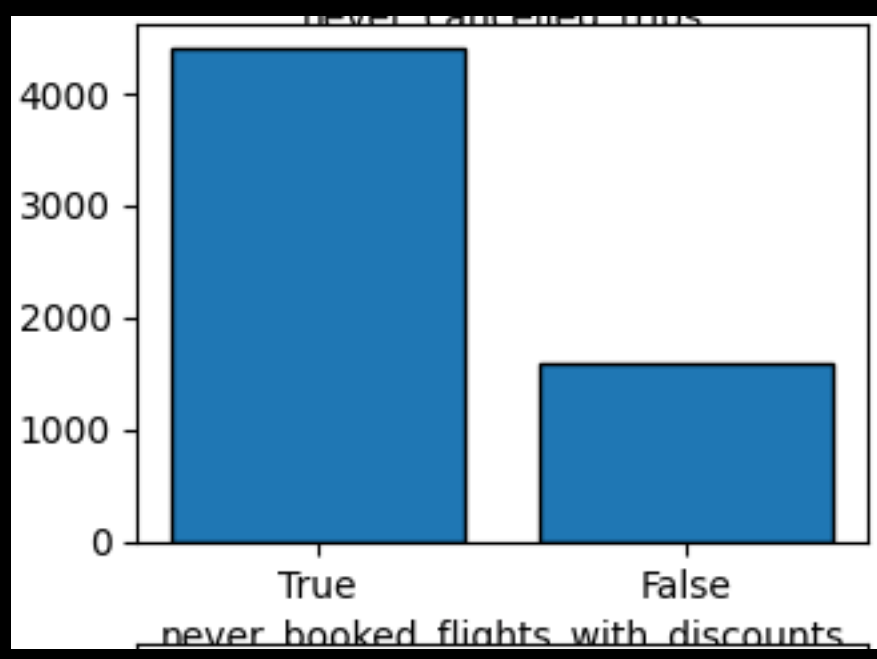
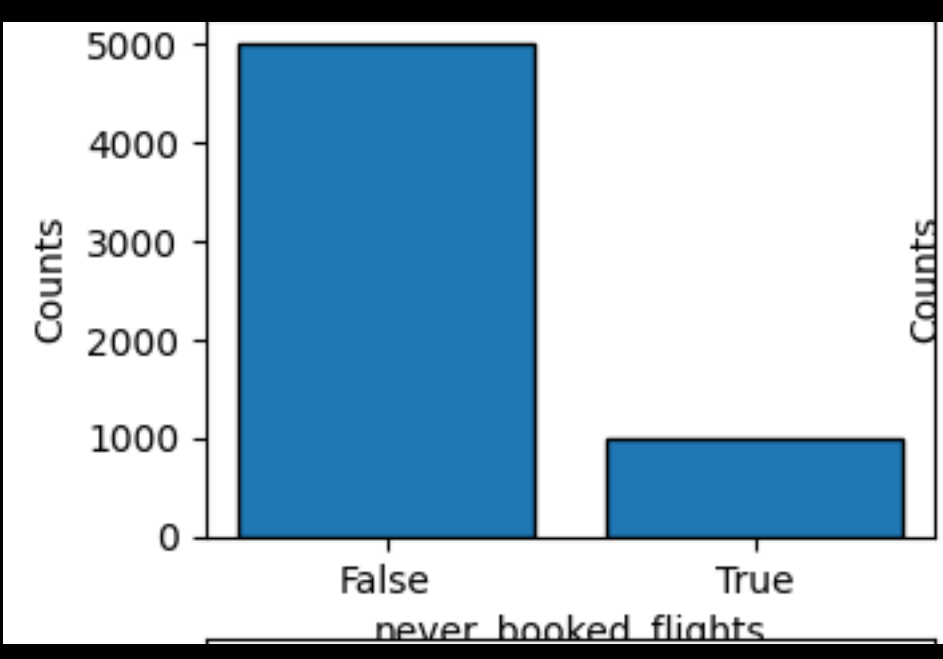
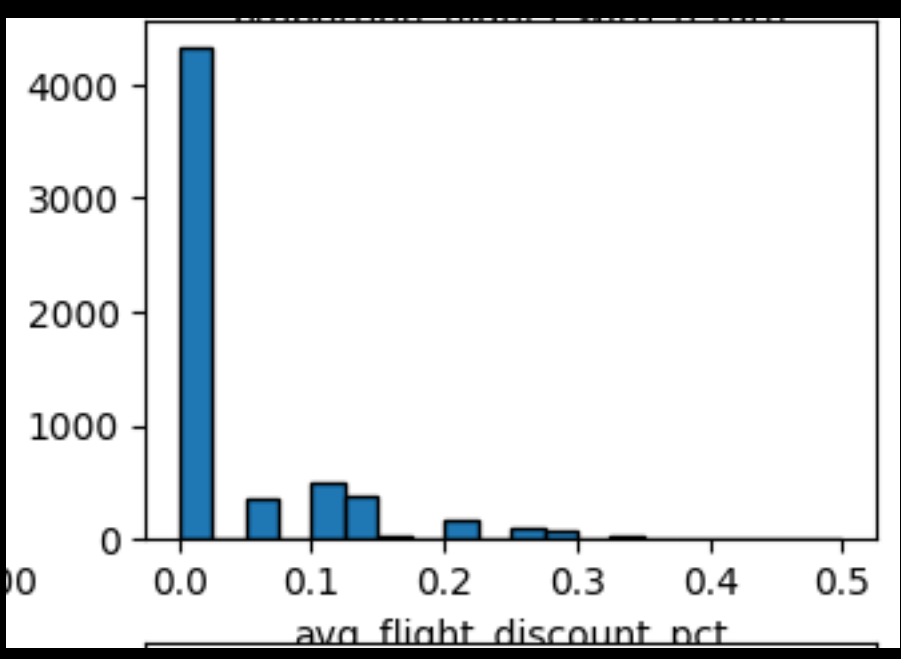
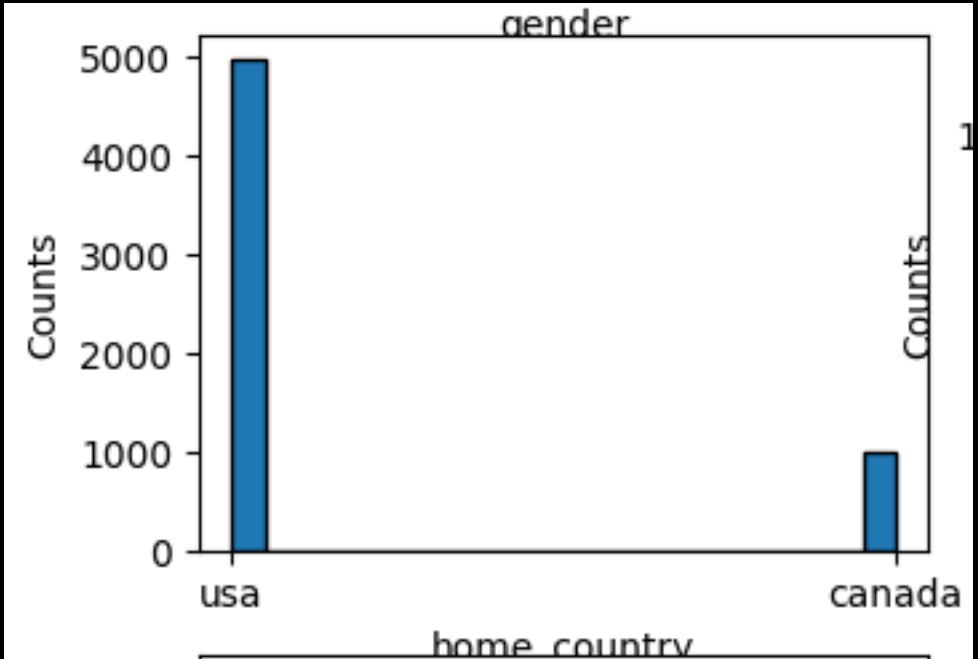
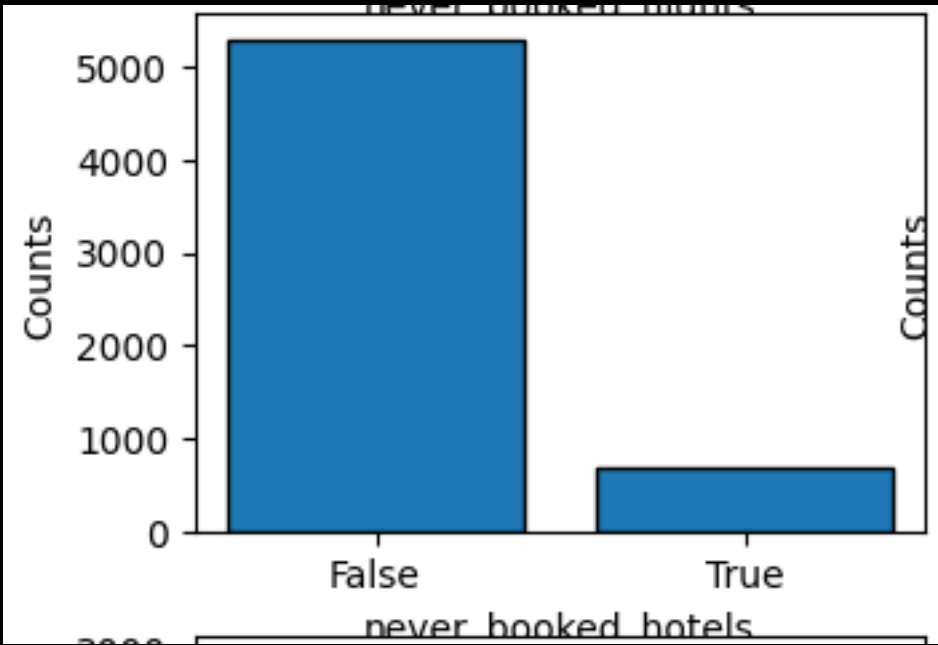
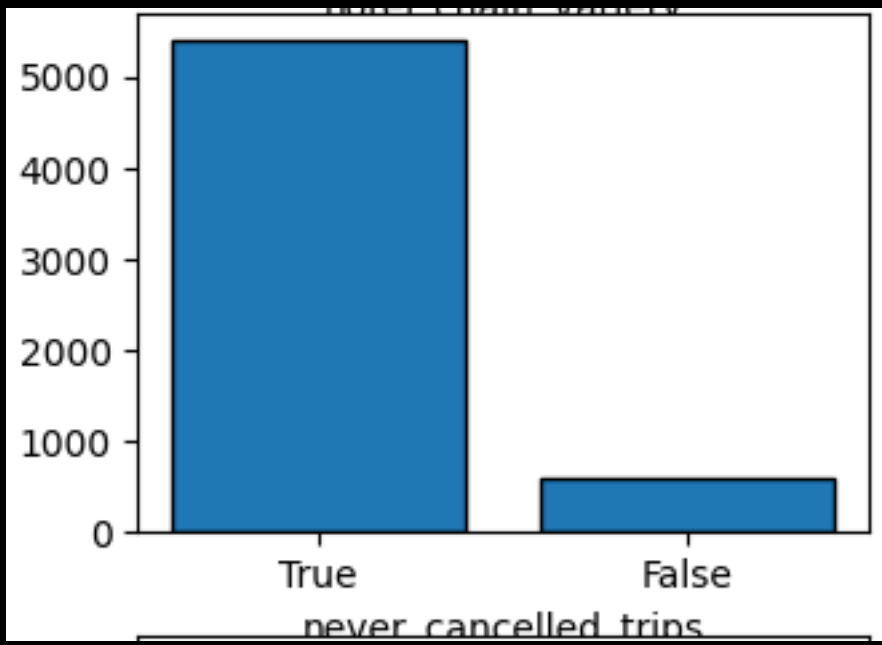
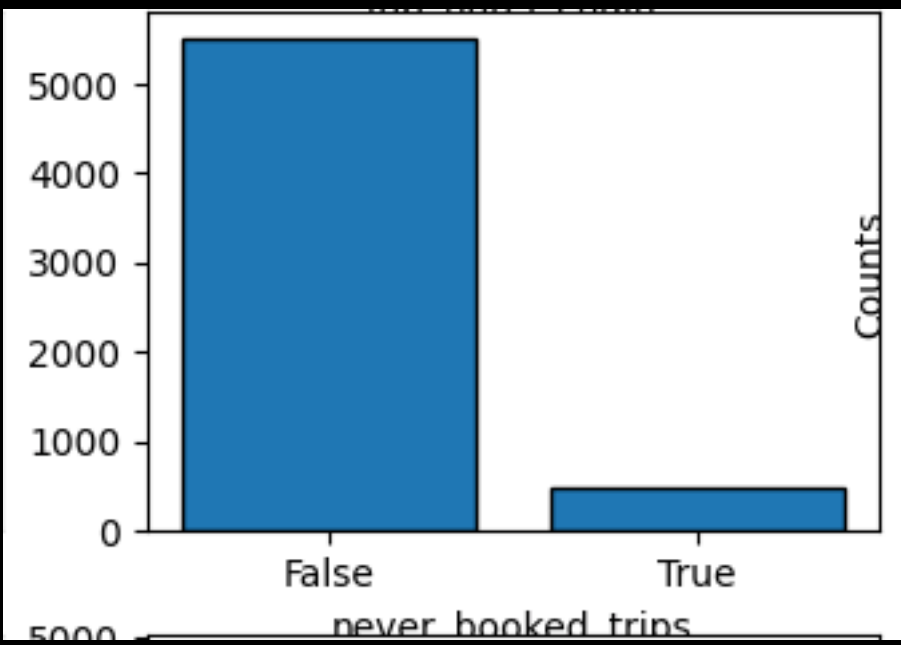
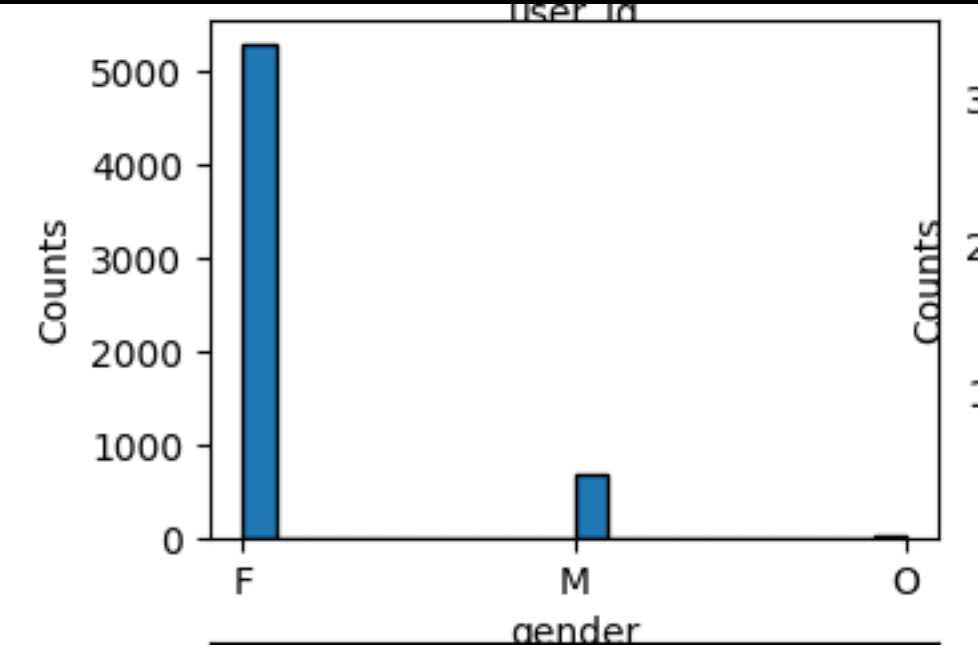
KMeans

col_0	count
group	
0	721
1	123
2	144
3	119
4	4723



- 5 groups - Elbow with highest silhouette score (0.72)
- Dominated by one class

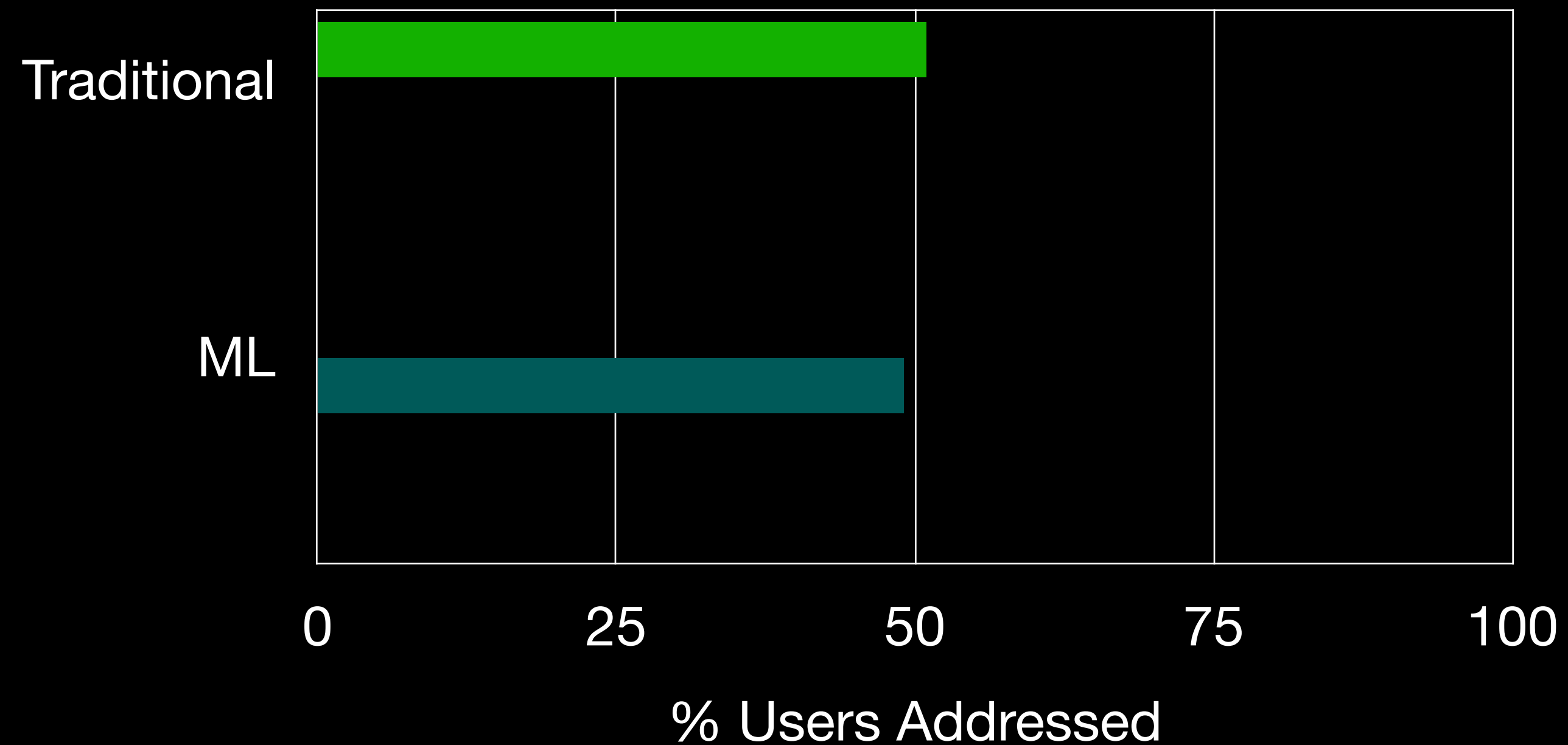
Homogenous Dataset



Hybrid Approach

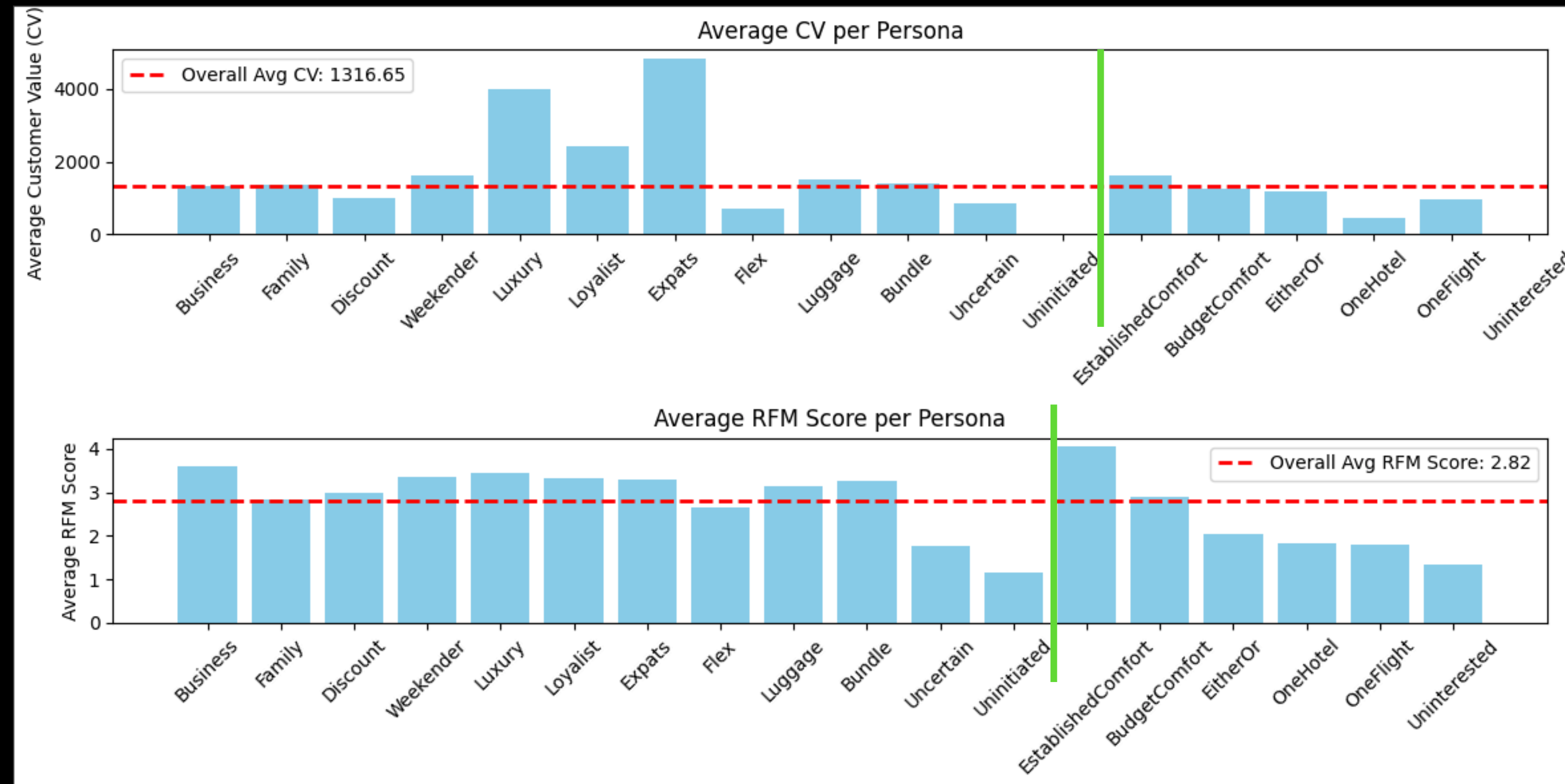
Traditional and ML Segmentation

Machine discovers the unaddressed



Traditional and ML Segmentation

Machine discovers the unaddressed



Persona Categorization

Strong - Growth - Weak

	Strong Perks based on motivations	Growth Perks based on upsell potential	Weak Lower priority
Traditional	Business Traveler Family Traveler Discount Hunter Weekender Couple Luxury Traveler Premium Loyalist Expatriate Flexibility-first Planner Luggage-heavy Frequent Flyer	Bundle Saver Uncertain Shopper Uninitiated	
ML	Established Comfort Seeker Budget Comfort Seeker	Bundle Saver Lite One-off Flyer	One-off Hotel Booker Uninterested

Strong Personas

Traditional Segmentation



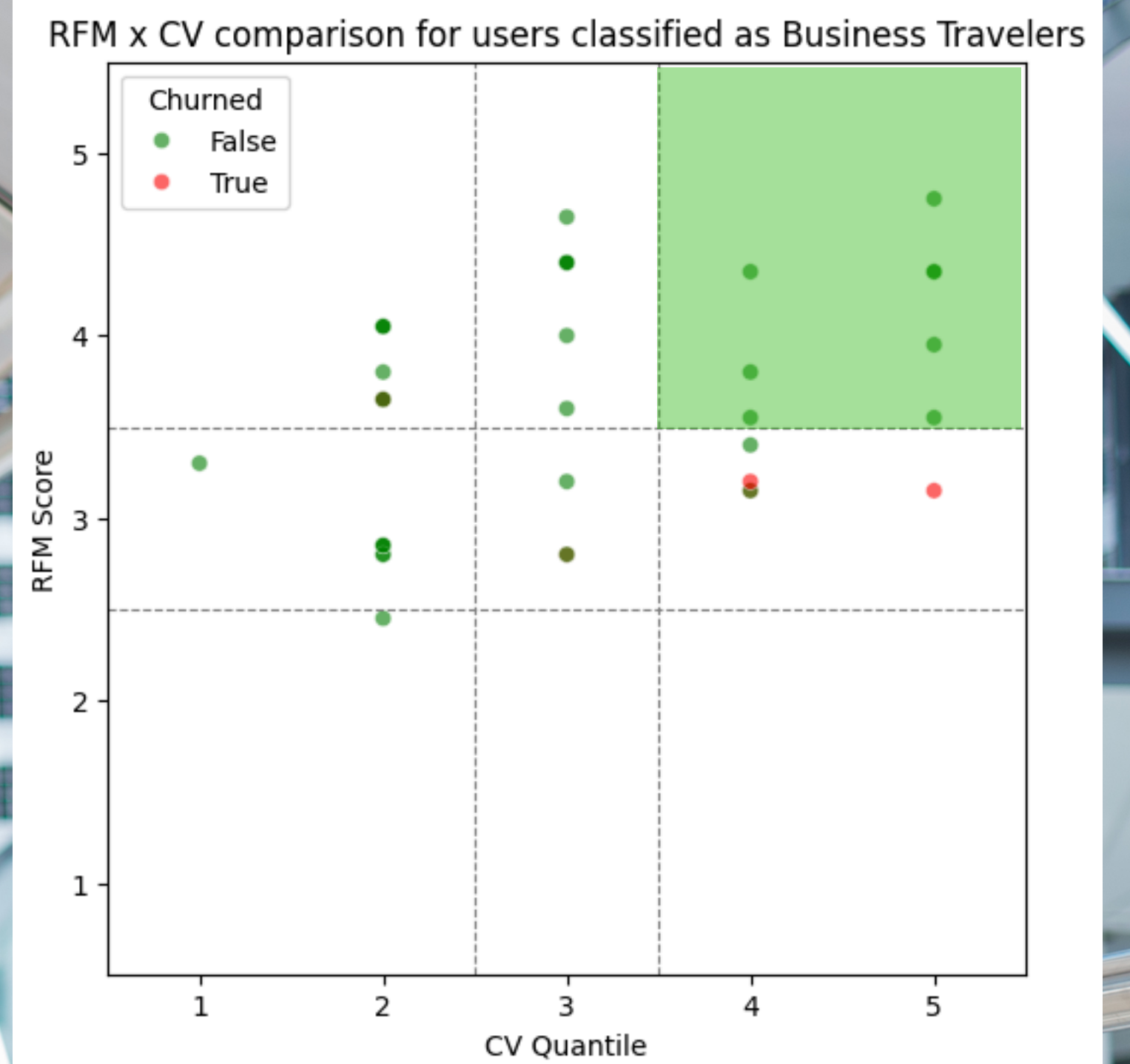
Numerous trips
Same destinations
Same airlines
Above average spending on flights
and hotels

Business Travelers

Flexibility and Comfort

Business Travelers

36 Users



VIP:

- Goal: retain
- Lounge access, loyalty bonuses, free cancellation

Core:

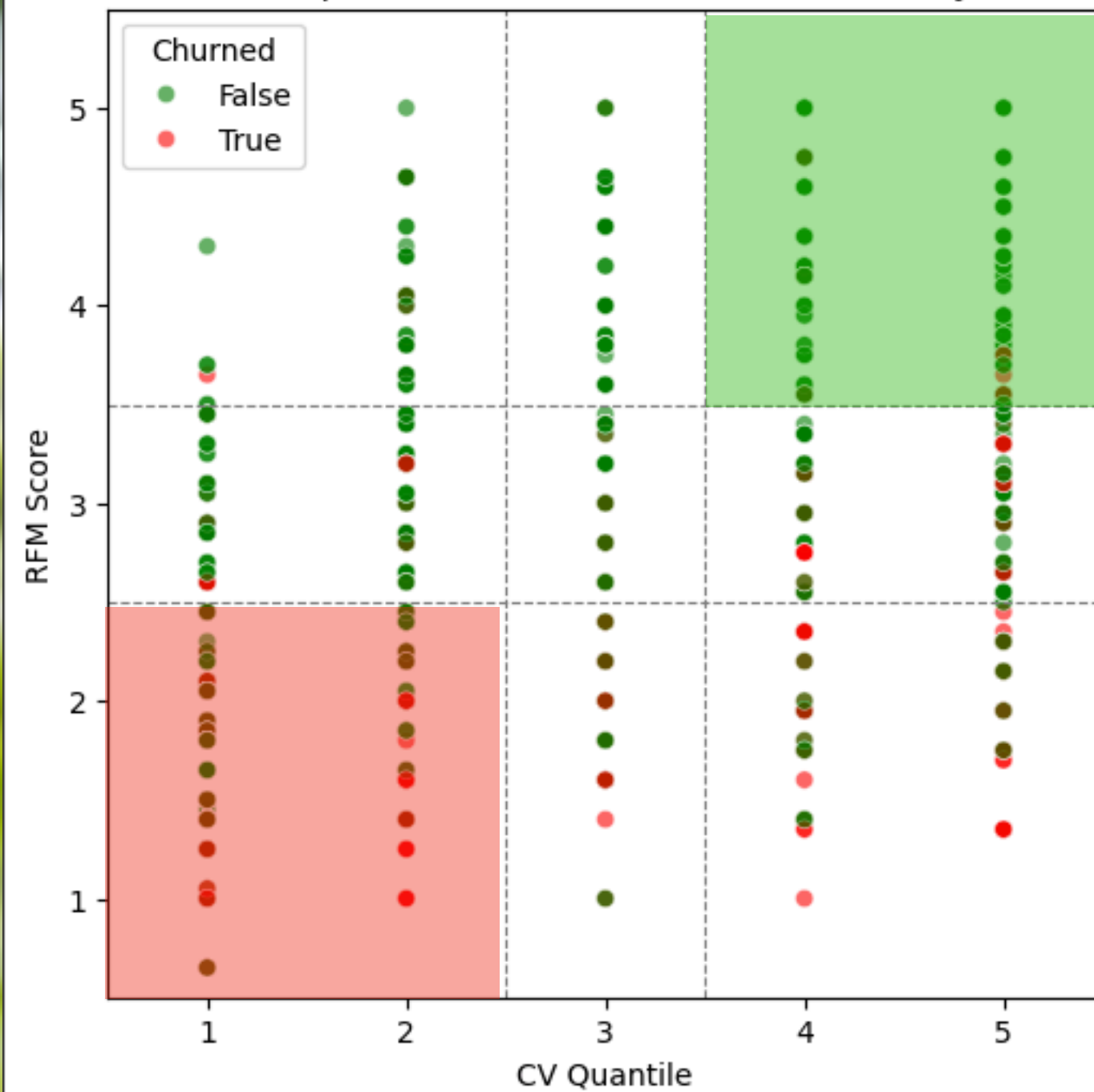
- Goal: improve spending
- Exclusive discounts to make more expensive options more attractive, free cancellation

Longer than average trip lengths
Has children or books two rooms or
more (because they have children)

Family Travelers

Cost-effective fun with kids

RFM x CV comparison for users classified as Family Travelers



VIP:

- Goal: retain / recover churned
- 1 Free hotel meal
- Family-friendly features e.g. early/late check-in/out, priority boarding, discount on airport-hotel transfers
- 1 night free hotel with flight (minimum X nights) for churned.

Core:

- Goal: nudge to higher levels
- Free hotel meal
- 1 night free hotel with flight (minimum X nights, Y seats, Z rooms)
- Upsell bundles (flight+hotel bundles at deeper discount, or more expensive bundles seats/hotels with bundled hotel-airport transfers or family activities)

At Risk:

- Goal: Low-cost activation
- Credit Wallet (e.g. 50 EUR off next booking)
- Free cancellation, price match guarantee.

Family Travelers

1040 Users

Lower average fare / room rate
Higher average clicks per session
because they are looking for deals

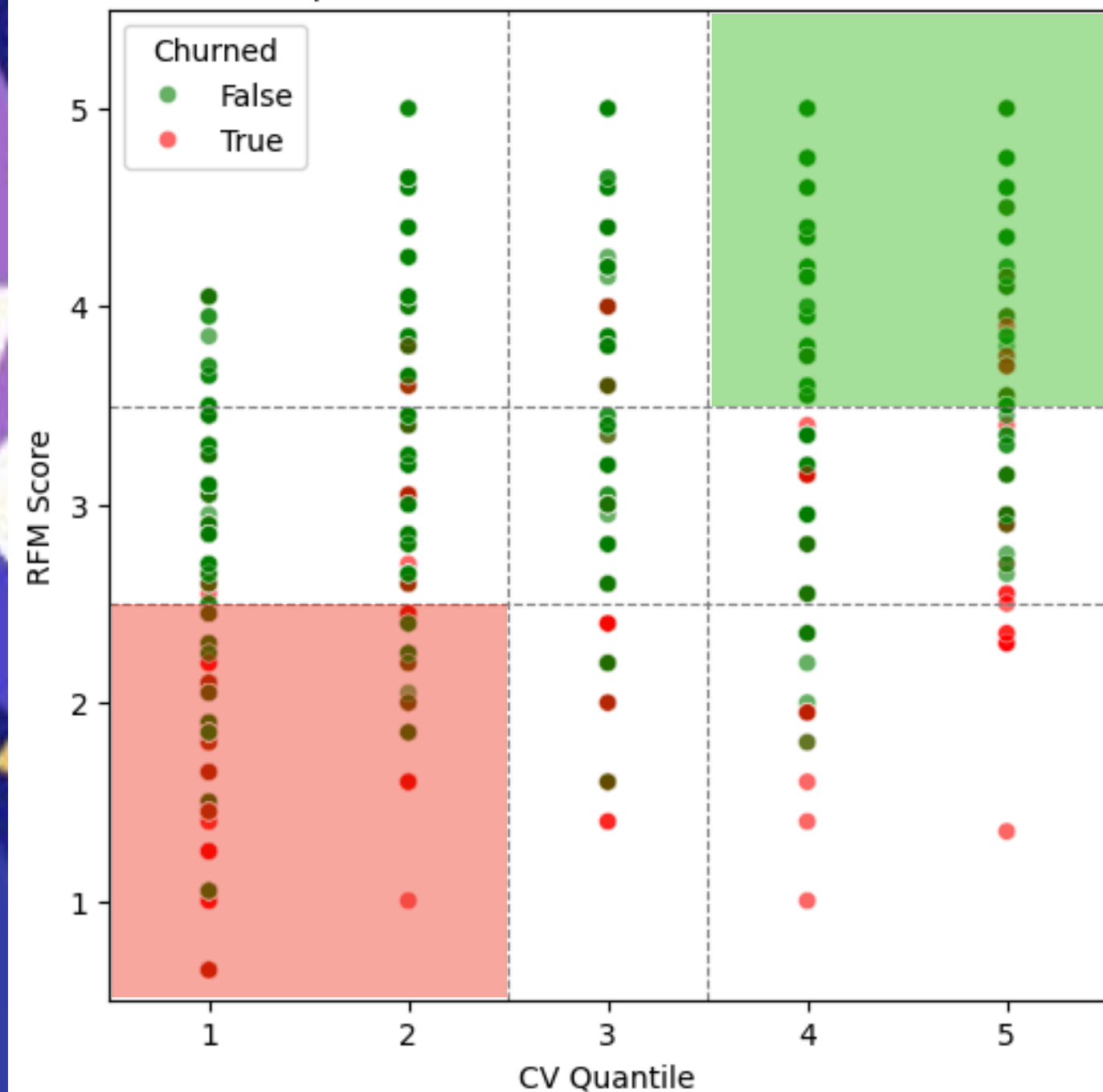
Discount Hunters

“Best Deal” perception

Discount Hunters

1114 Users

RFM x CV comparison for users classified as Discount Hunters



VIP:

- Goal: sustain loyalty with minimal cost
- Premium deal bundles at discounted rates

Core:

- Goal: improve activity i.e. volume over luxury
- Flash sales, price match guarantee

At Risk:

- Goal: Low-cost activation
- Personalized deal alerts / reminder emails

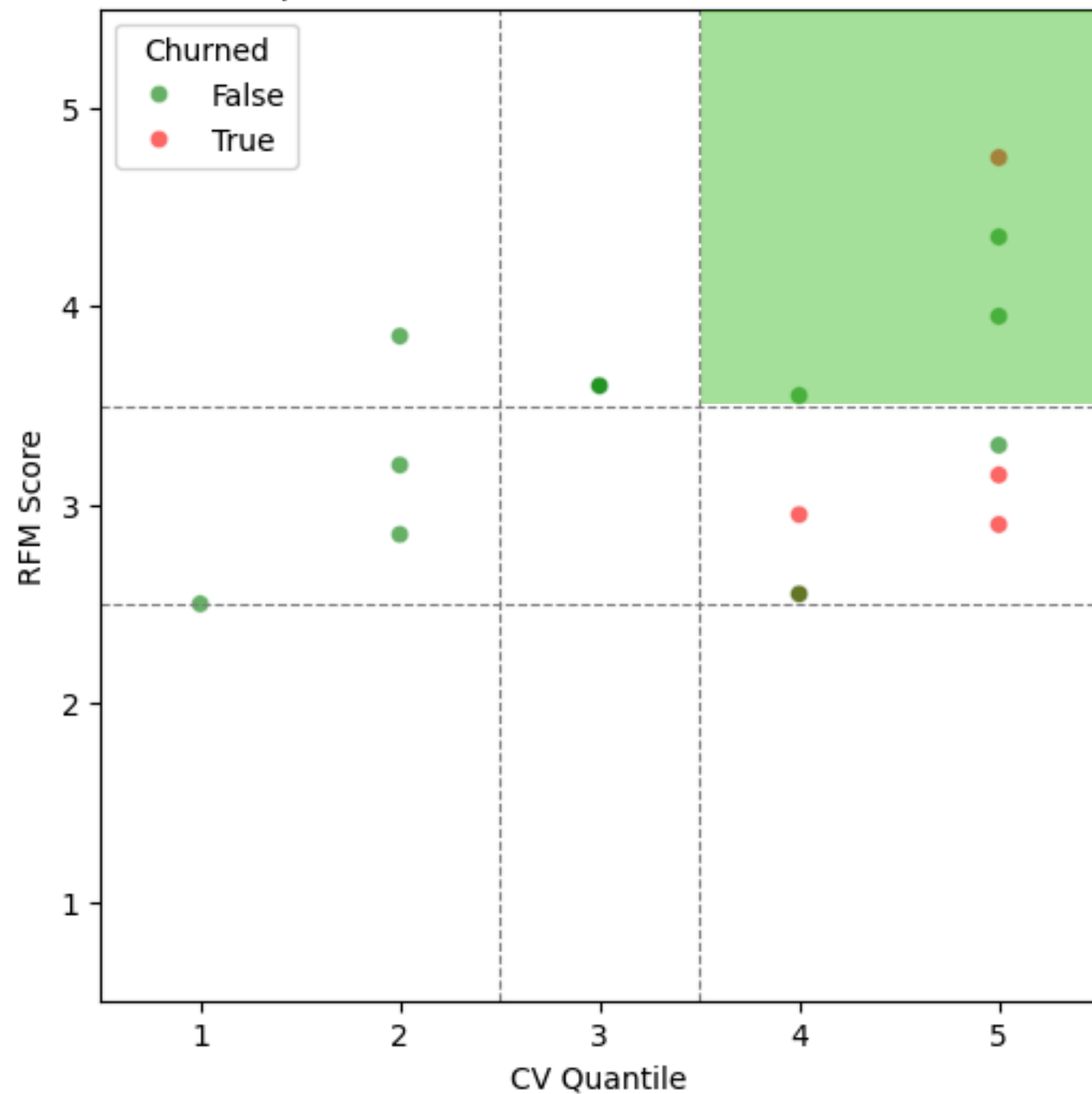
A romantic couple is seen from behind, sitting in a hammock strung between two large palm trees. The scene is set at sunset, with a warm, golden-orange glow in the sky. The couple is silhouetted against the bright light, creating a peaceful and intimate atmosphere. The hammock is made of a patterned fabric and is supported by wooden poles. The palm trees are large and have long, feathery fronds that hang down. The overall mood is relaxing and romantic.

Books 1 room, 2 seats.
Shorter trip duration
High variety of destinations
Shorter booking lead time

Weekender Couples

Quick experiences

RFM x CV comparison for users classified as Weekender Couples



VIP:

- Goal: reward and retain
- VIP Weekend Packages (e.g. room upgrade, late check-out)
- Free hotel breakfast

Core:

- Goal: improve spending / frequency
- “Next Trip” discount voucher
- Free couple activity (e.g. spa, bike rental) on 2nd trip in a period (e.g. month)

Weekender Couples

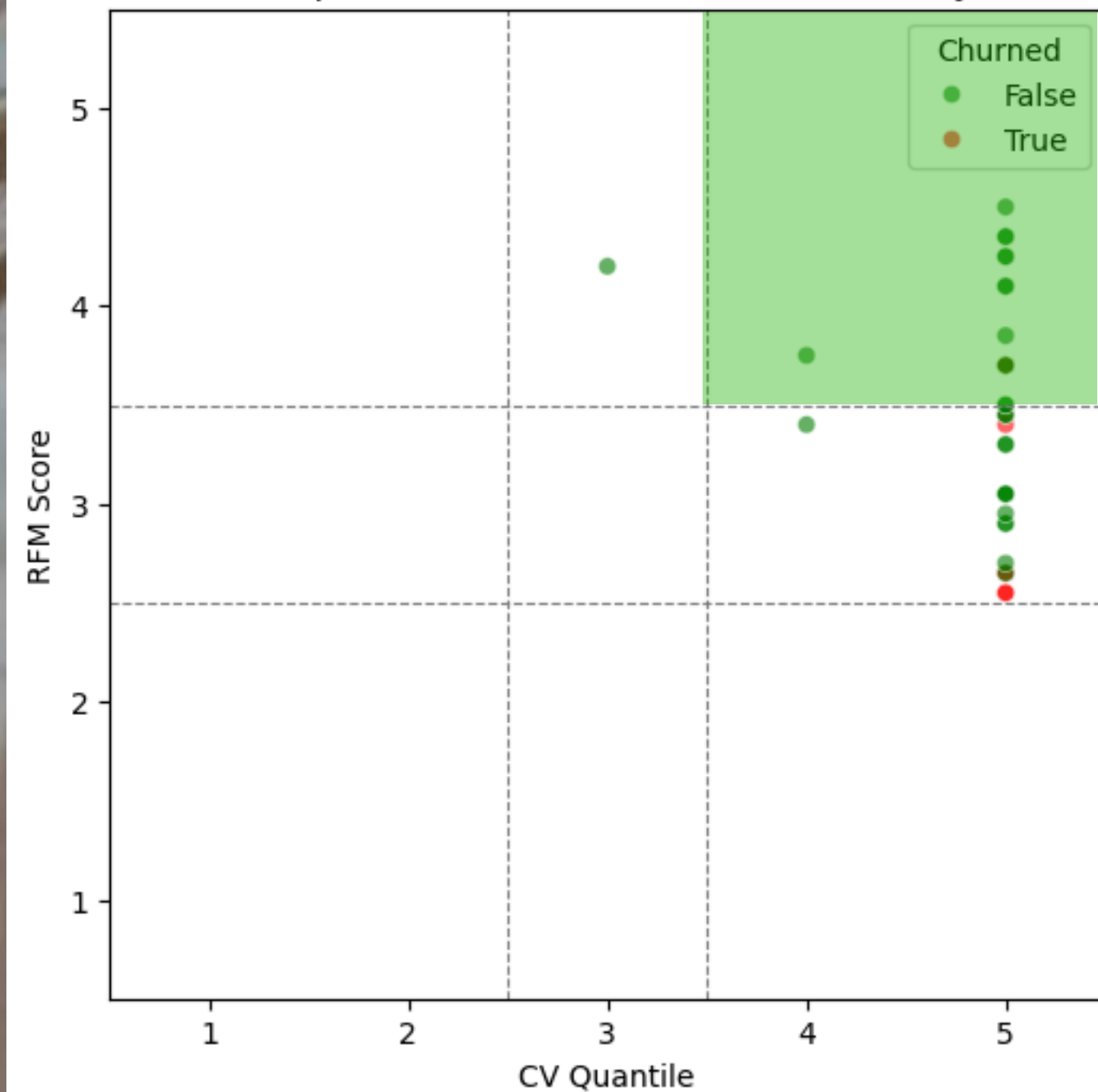
16 Users

Booked more than one trip
High average flight and hotel price
High total spend
Low discount use
Low average clicks because they
are not too concerned about finding
the best deals.

Luxury Travelers

Exclusivity with no Friction

RFM x CV comparison for users classified as Luxury Travellers



VIP:

- Goal: reward and retain
- Invite-only elite status
- Dedicated concierge, chauffeur for transfers, private gifts at hotel.

Core:

- Goal: cement loyalty, improve consistency
- Dedicated concierge, guaranteed suite upgrade, exclusive event invites

Luxury Travelers

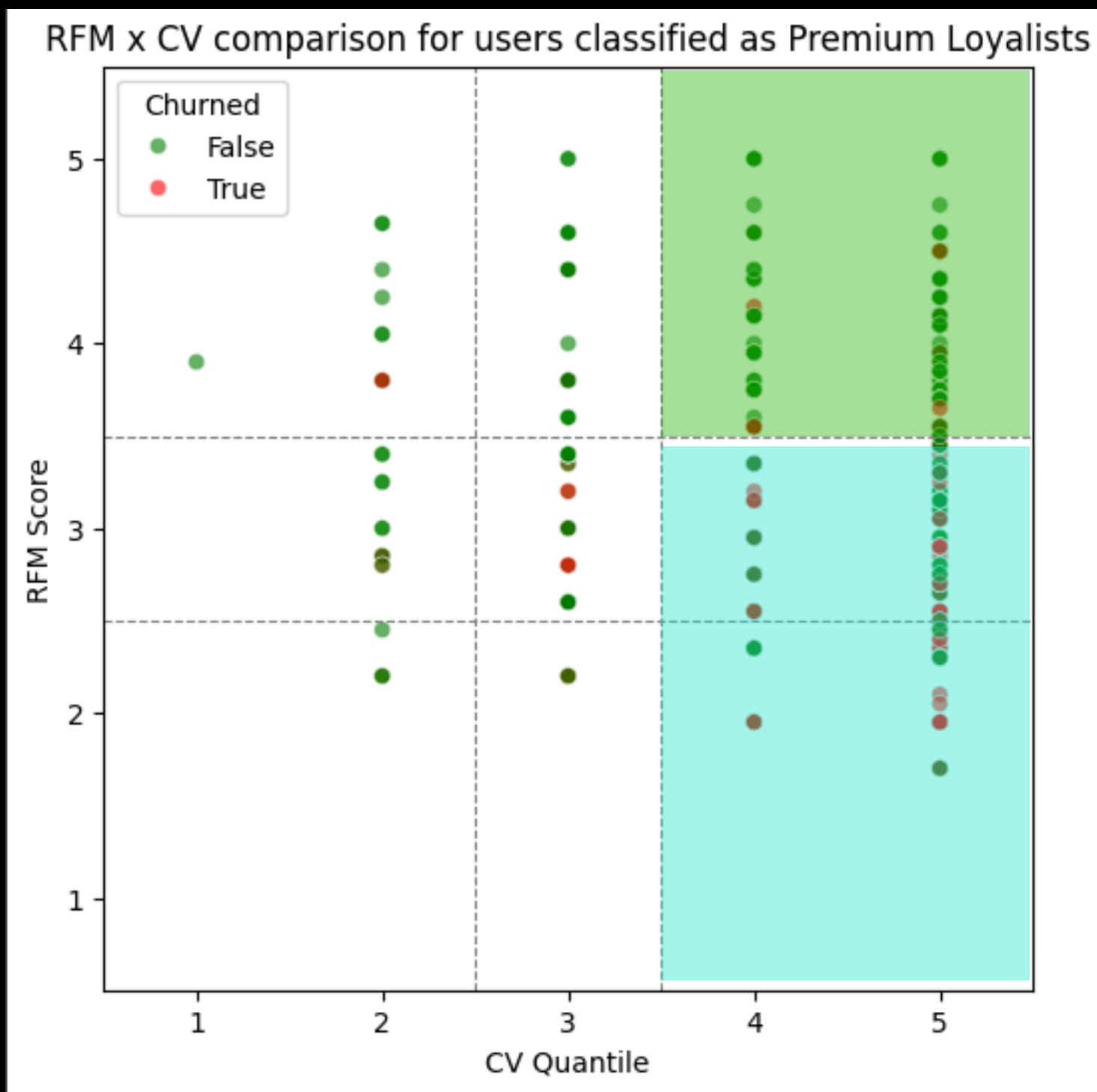
35 Users

Higher total spend
Higher average flight / hotel price
Loyal to airline and/or hotel chain
Low trip cancellation rate



Premium Loyalists

Reward brand loyalty



VIP:

- Goal: cement top loyalty
- Tie-up with loyalty program providers to upgrade status (e.g. gold to platinum / platinum to invite-only tier)

At Risk:

- Goal: Reactivate churned users, promote more consistency
- Temporary reinstatement of premium status for certain time period (e.g. reactivated Gold status for next 3 months, book twice to keep until end of year)

Premium Loyalists

434 Users



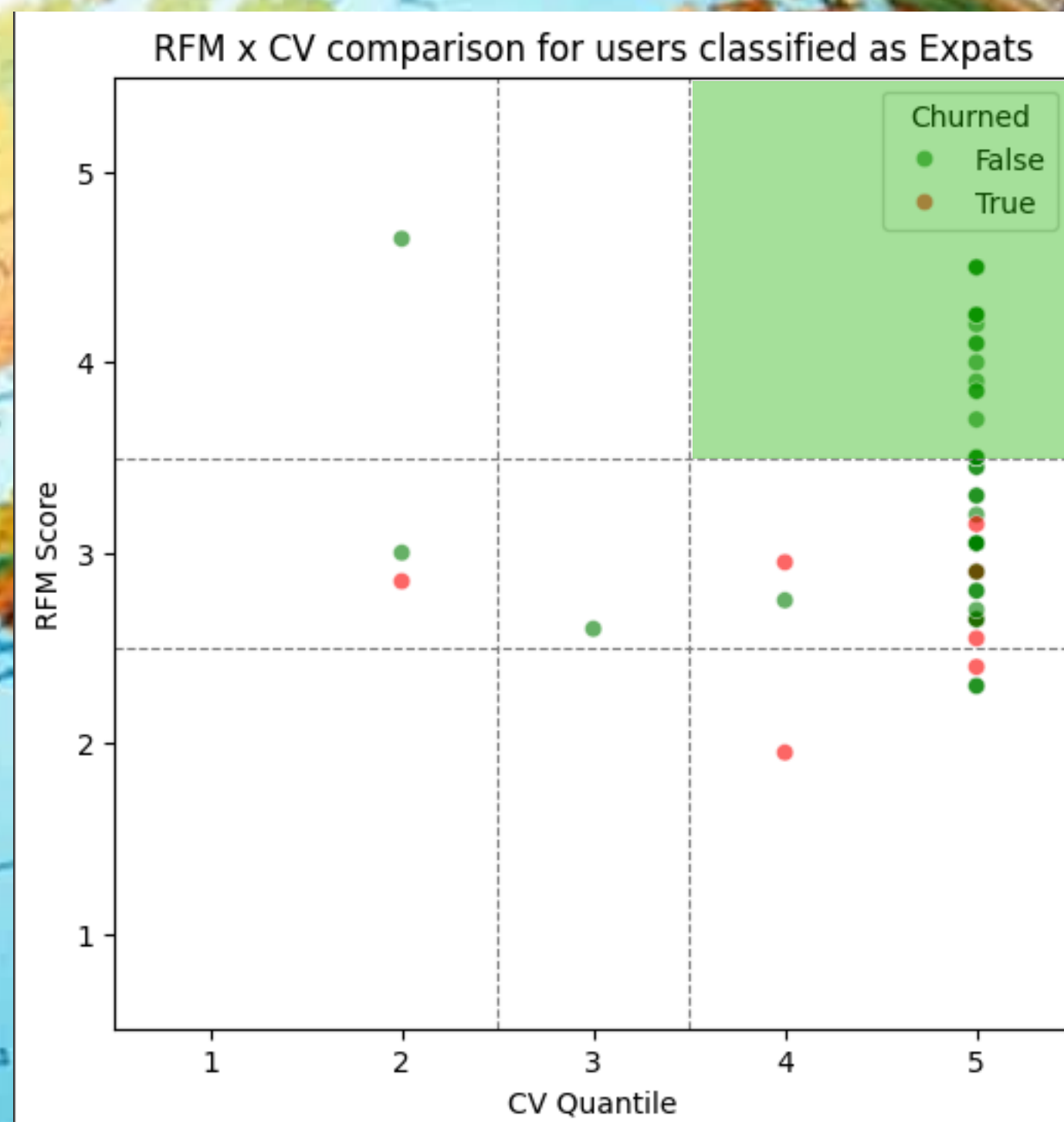
High proportion of long-haul flights
Low flight route variance
Longer trip duration
Regularly travels with checked bags
Longer booking lead time

Expats

Cheap and comfy ride home

VIP:

- Goal: cement top loyalty
- Get more value from their “miles flown” e.g. Business Class meal or legroom upgrade for long-haul flight.
- Early bird booking discounts
- Free checked bag



Core:

- Goal: recover from churn, improve consistency
- Early bird booking discounts
- Free checked bag

Expats

59 Users

A man with short dark hair, wearing a blue tank top and black shorts, is performing a yoga pose on a dark mat in a grassy field. He is in a side plank variation, with his left leg bent and his right leg extended. His hands are clasped together near his right knee. The background is a lush green lawn with some white flowers in the distance.

Higher average booking lead time
Non-zero cancelled trips
Fairly active on the platform

Flexibility-First Planners

Security in Uncertainty

Flexibility-First Planners

432 Users

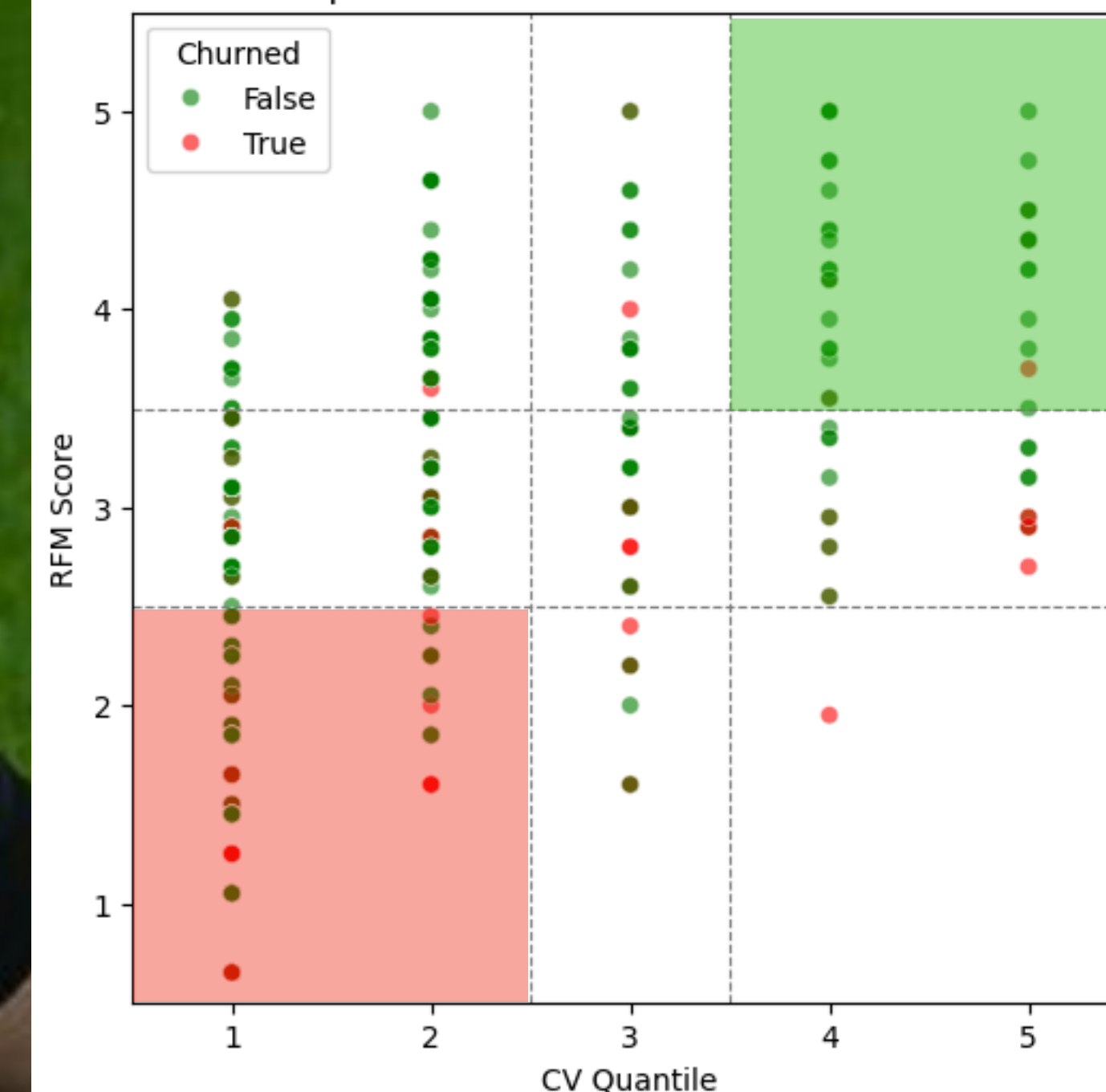
Core:

- Goal: try to move to higher tier
- Free cancellation within 24-48hrs, free travel insurances if booking higher cost flights or hotels, or as perk for Nth booking in a period.

VIP:

- Goal: provide maximum sense of security
- Free cancellation, inclusion of travel insurances

RFM x CV comparison for users classified as Flex-first Planners



At Risk:

- Goal: Low-cost activation / deprioritize
- Deal alerts / reminder emails



Had booked at least two flights
Checked at least one bag on
average

Luggage-heavy Frequent Flyers

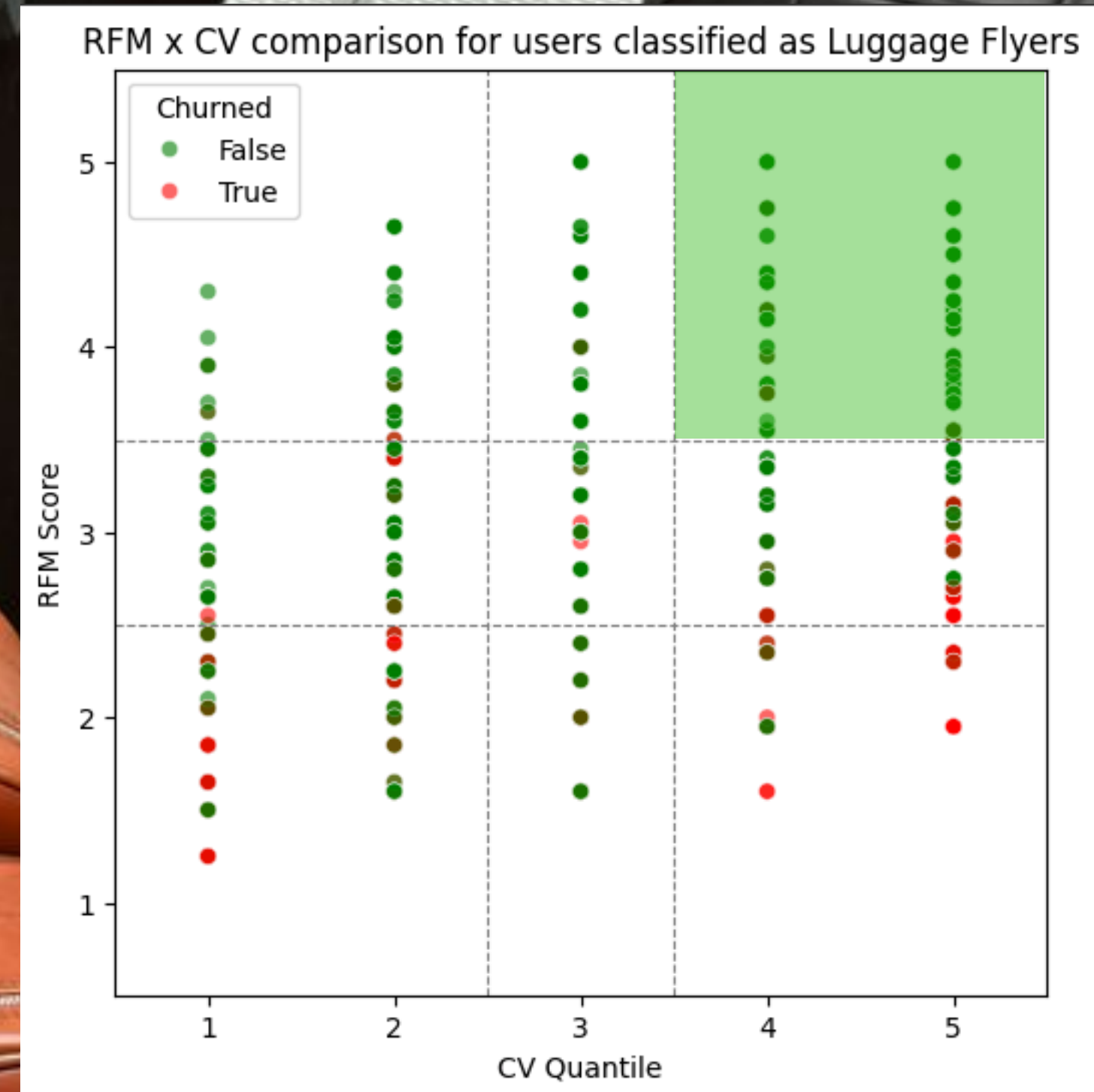
They don't travel light

VIP:

- Goal: retain and maintain
- Free checked bag / Free hand carry capacity increase
- Home-Airport-Hotel transport assistance

Core:

- Goal: Encourage higher-value trips
- Already booked trips at minimum 50th percentile, so frequency is not the issue.
- Free checked bag for minimum X bags checked. (e.g. buy 2 get 1)
- Free hand carry capacity increase for higher tier trips / for bundle bookings.



Luggage-heavy Frequent Flyers

899 Users

Personas for Cultivation

Traditional Segmentation

A large bundle of cinnamon sticks, tied with dried banana leaves, is shown against a grey background. The bundle is rectangular and composed of many individual cinnamon sticks. The ties are made of dried banana leaves, which are light brown and have a fibrous texture. The cinnamon sticks are a warm, reddish-brown color.

Lower than average proportion of
trips with both flights and hotels
booked

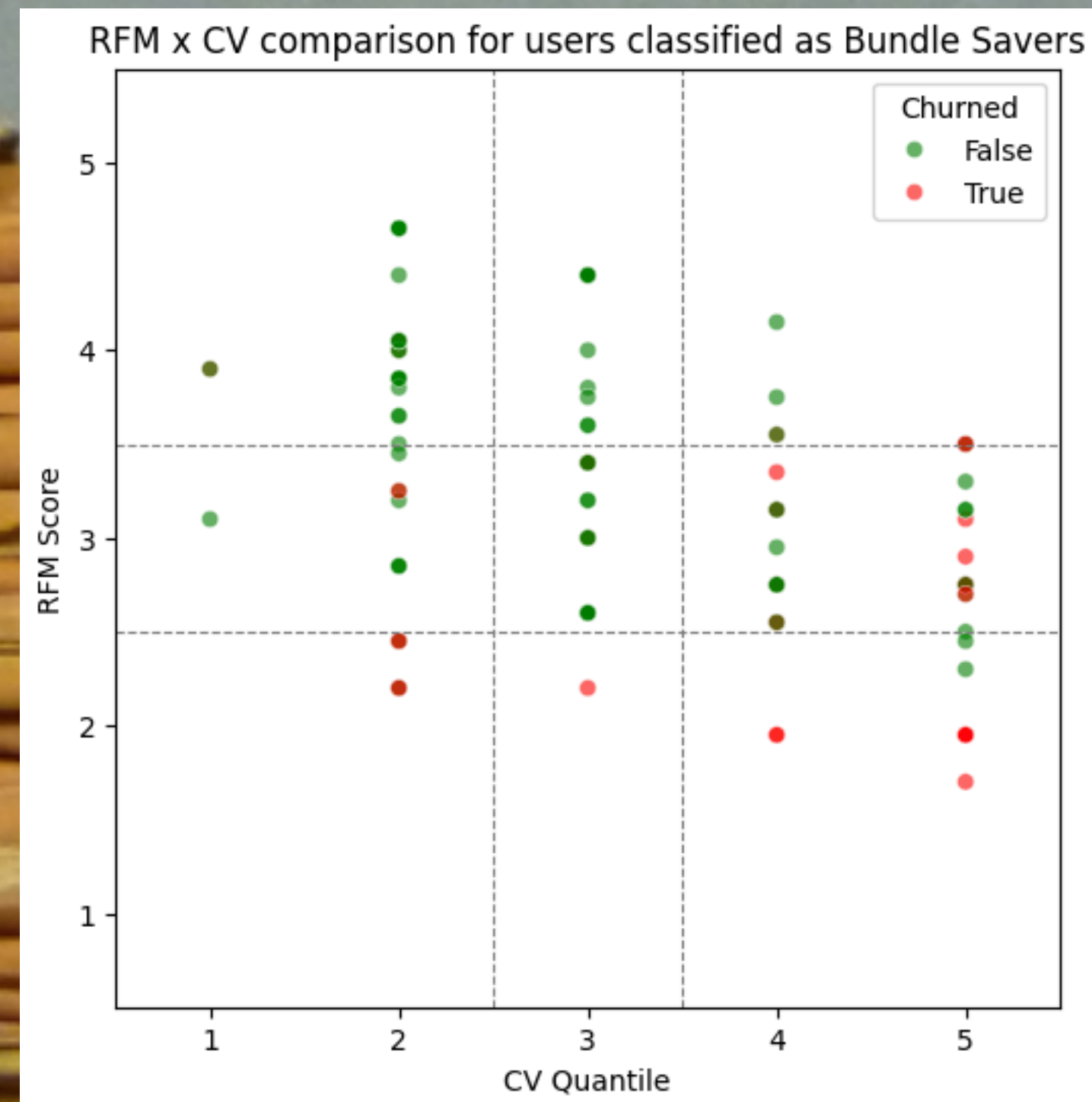
Short average trip length
Mid-high total spend, but not
extravagant

Bundle Savers

Try both flights and hotels

Bundle Savers

110 Users



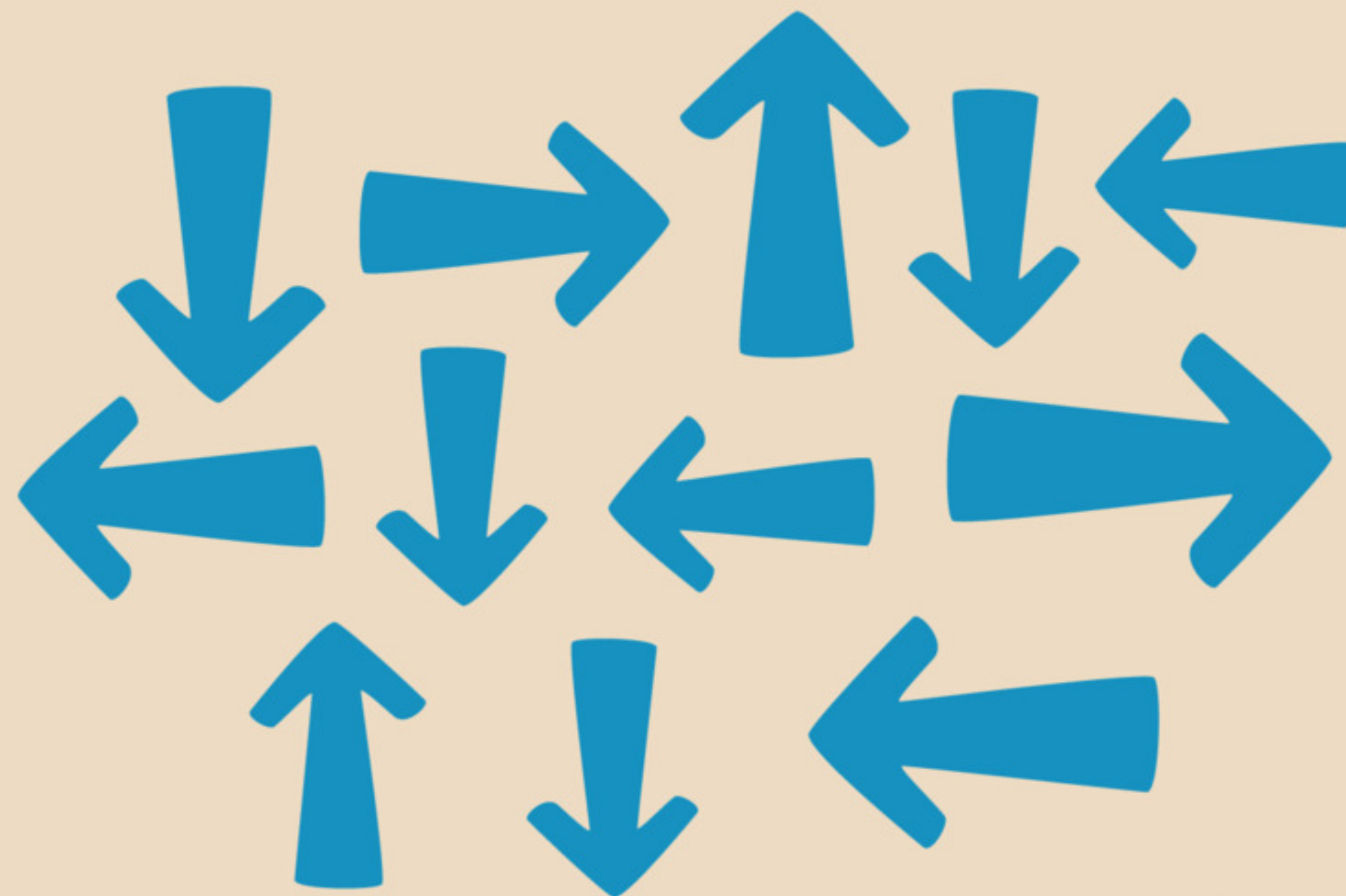
Lower than average proportion of trips with both flights and hotels booked

Short average trip length

Mid-high total spend, but not extravagant

Perk: 1 free night with flight (minimum X nights at mid-high tier hotel, roundtrip)

- 1 extra night is meaningful for shorter trip length
- Decent hotel more attractive than dorm/hostel



Booked only one trip
Fairly active on the platform
Spends a lot of time and clicks a lot
every session

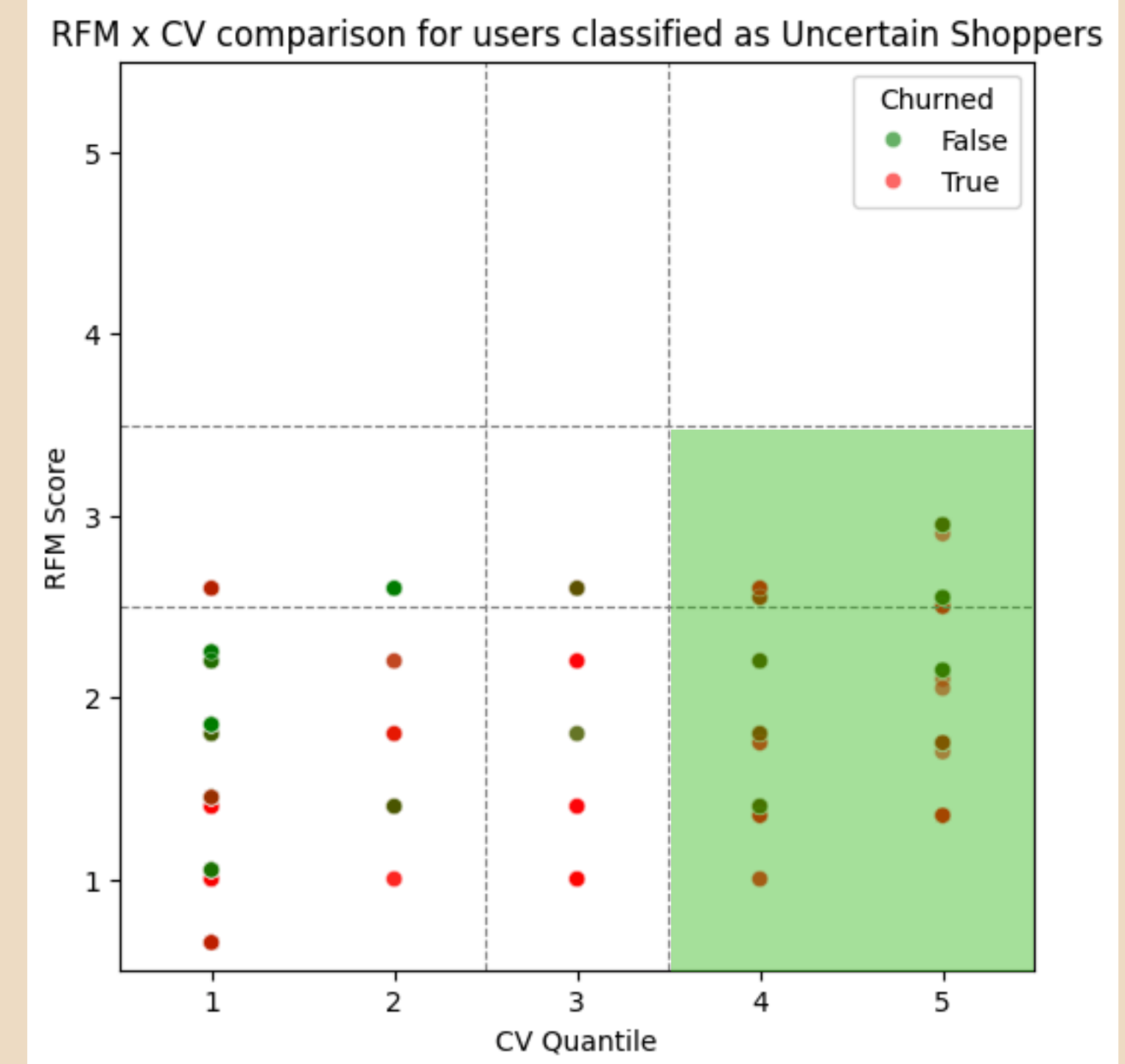


Uncertain Shoppers

Nudge them to decide

Uncertain Shoppers

110 Users



Growth Potential

- Goal: Reactivate because most have churned
- Free cancellation, price match guarantee, personalized emails with price comparisons and summaries for routes they research

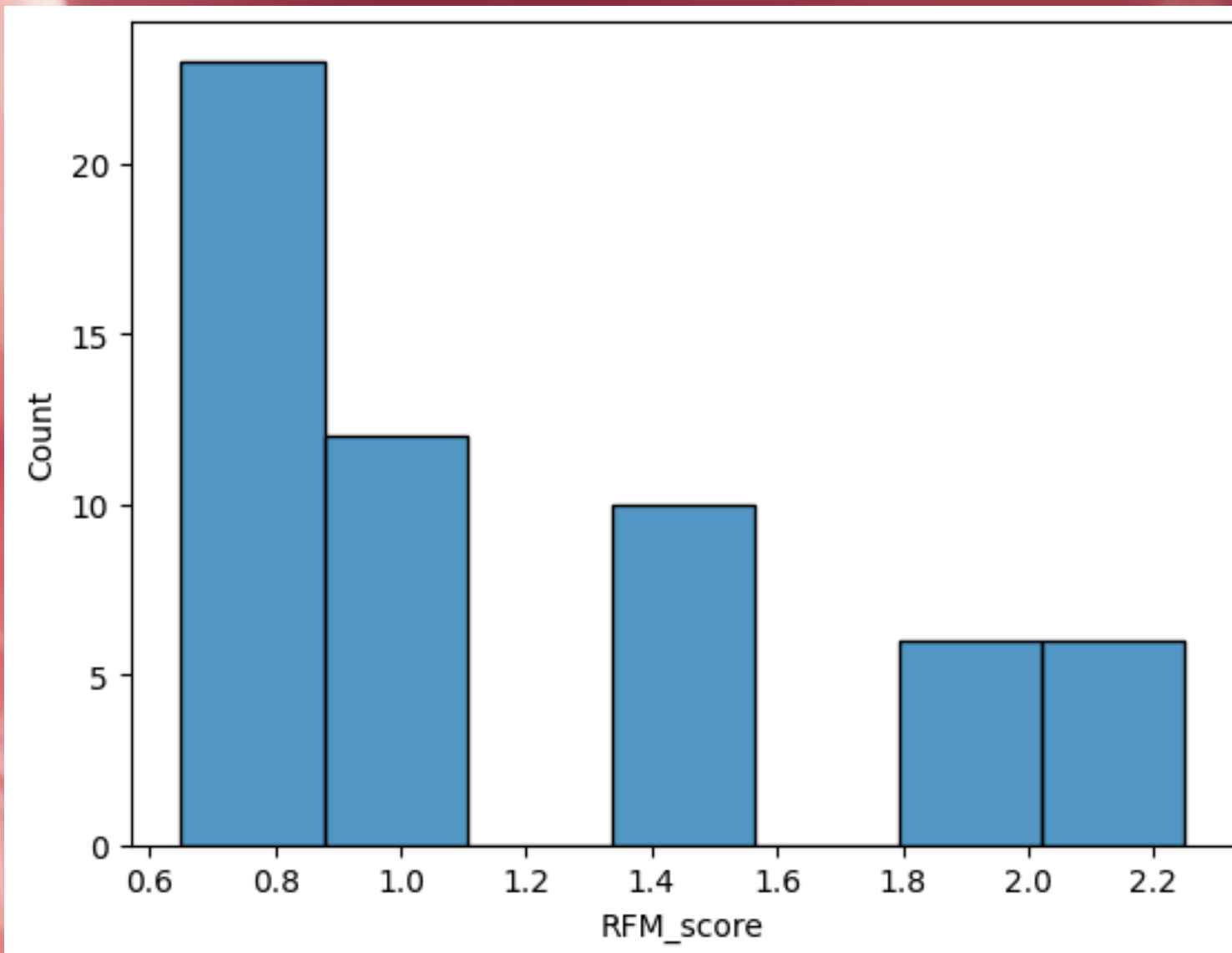


Never booked a trip before
High average session length and
clicks per session

The Uninitiated

The first hit is for “free”

Never booked a trip before
High average session length and
clicks per session



Perk: 1\$ flight and 1\$ for one night
(with minimum number of nights)
- Overcome the “first purchase” hurdle to
make future purchases easier on the
psyche.

The Uninitiated

57 Users

Personas found by the Machine

Strong Personas

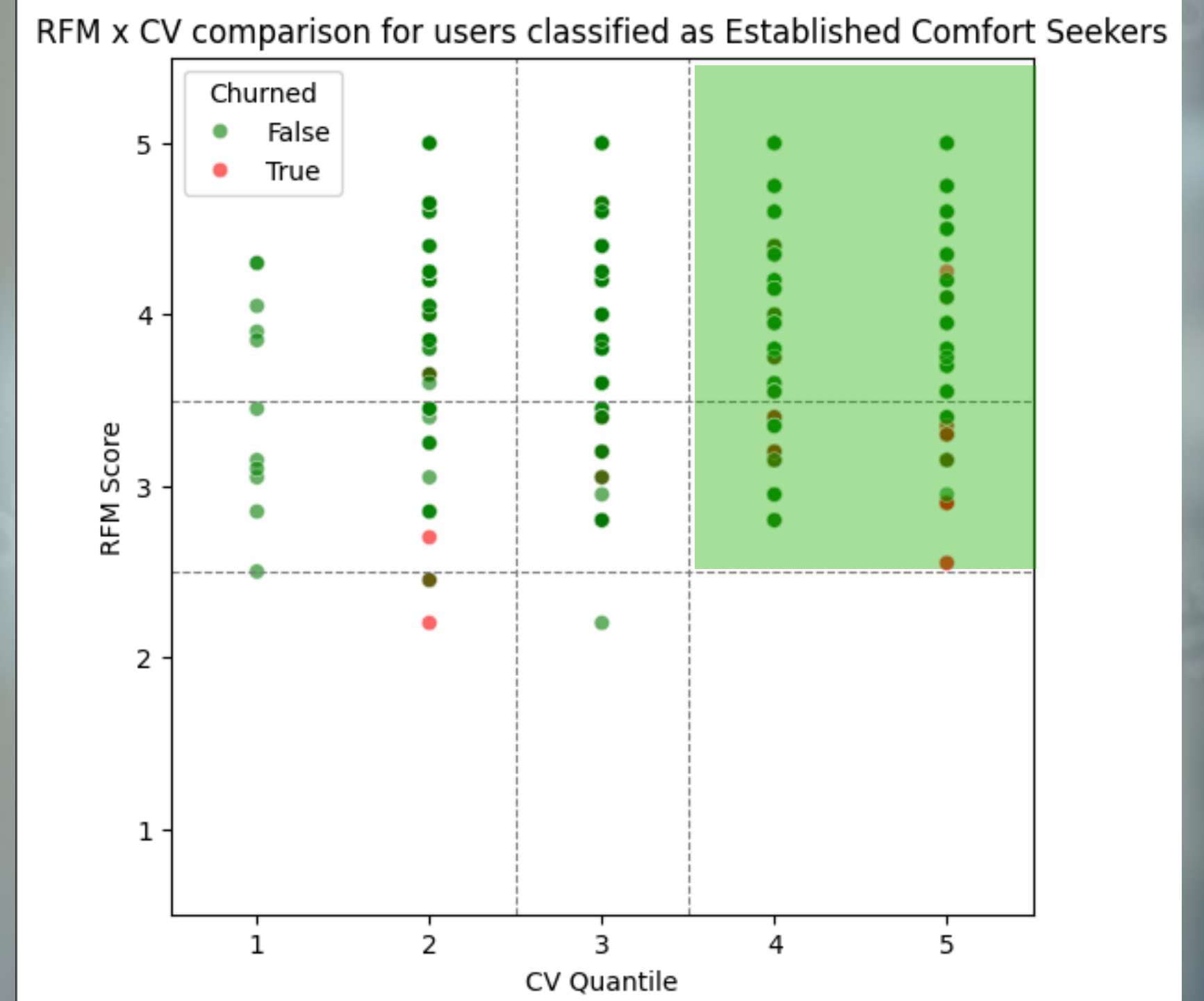
Established Comfort-Seeker

Enjoying life

Normally distributed around 42.5 years old
Slightly lower price point than luxury
travelers, and also does a little more
research (higher clicks)
Often books hotel and flights together.

Established Comfort-Seeker

746 Users



VIP:

- Goal: retain and maintain
- Offer free/discounted experiences together with the trip bundles that are fitting for middle-aged people (e.g. spa, boat, museum, wine-tasting)

Core:

- Pretty high RFM across the board
- Goal: try to move to higher Customer Value
- Exclusive discounts on higher-tier destinations with bundled experiences.

The background image is a collage. On the right, a man with short brown hair and a light beard, wearing a light blue button-down shirt and a black backpack, is smiling at the camera while holding a bowl of food and a glass of beer. On the left, there is a close-up of a white bowl filled with a meal consisting of fried chicken, shredded yellow cheese, green onions, and other ingredients. A yellow text box is overlaid on the top left of the food image.

Lower-budget and fewer trips mirror of
Established Comfort Seekers
Centering around 41.9 years old
Also books mostly flights and hotels
together

Similar strategy with Established Comfort
seekers, just at a lower price point.

Budget Comfort-Seeker

Enjoying life on a budget

VIP:

- Goal: retain and maintain
- Offer free/discounted experiences together with the trip bundles that are fitting for middle-aged people (e.g. spa, boat, museum, wine-tasting). Free hotel meal

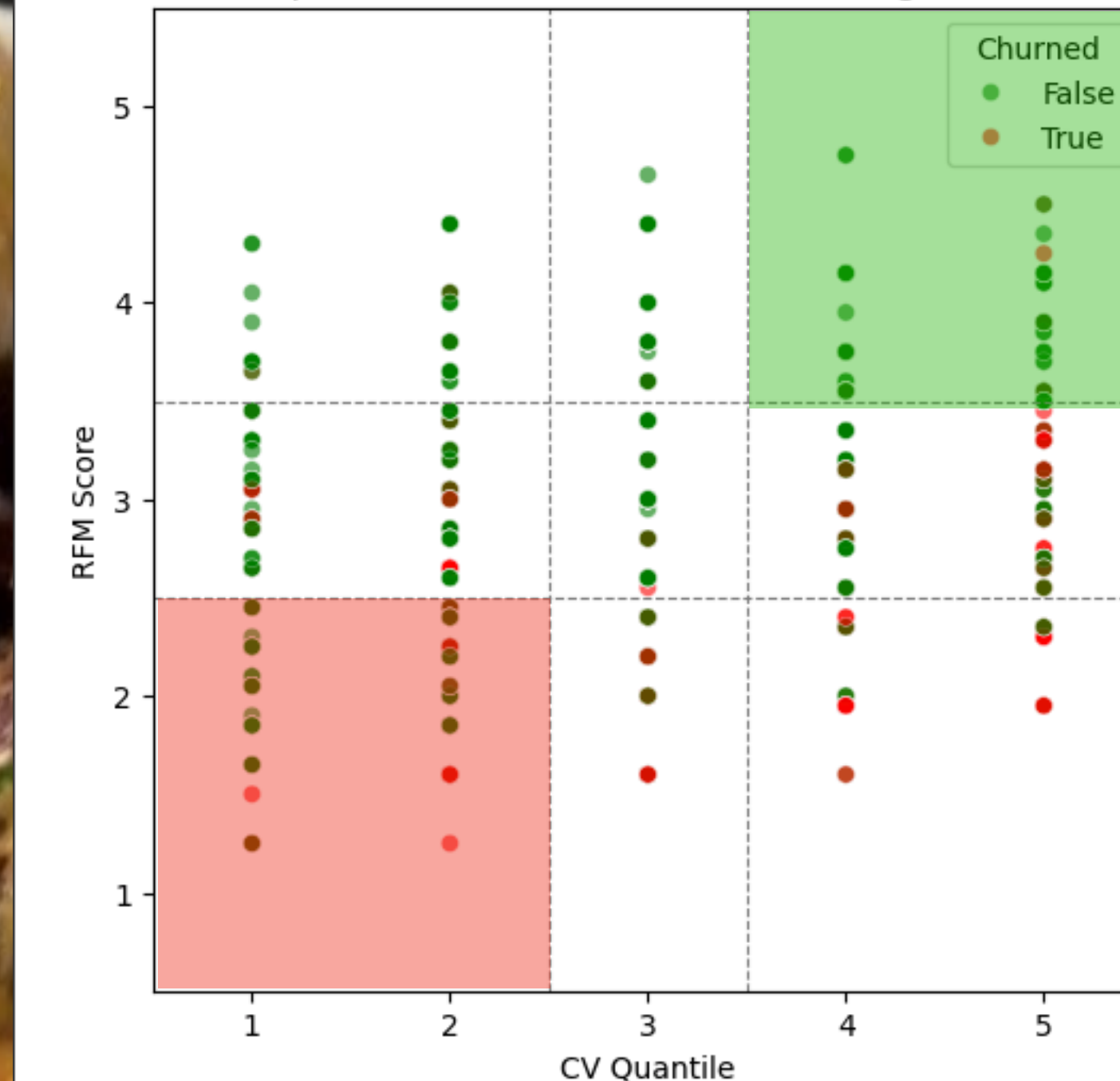
Core:

- Goal: try to move to higher Customer Value
- Exclusive discounts on higher-tier destinations with bundled experiences. Free hotel meal

At Risk:

- Goal: Low-cost activation / deprioritize
- Deal alerts / reminder emails

RFM x CV comparison for users classified as Budget Comfort Seekers



Budget Comfort-Seeker

966 Users

Personas found by the Machine

For Cultivation



Bundle Savers Lite

Try both flights and hotels

Lower Recency/Frequency mirror of
Bundle Savers but with longer
average trip length
Often books only either hotel or
flight
Some discount sensitivity
Potential for upsell

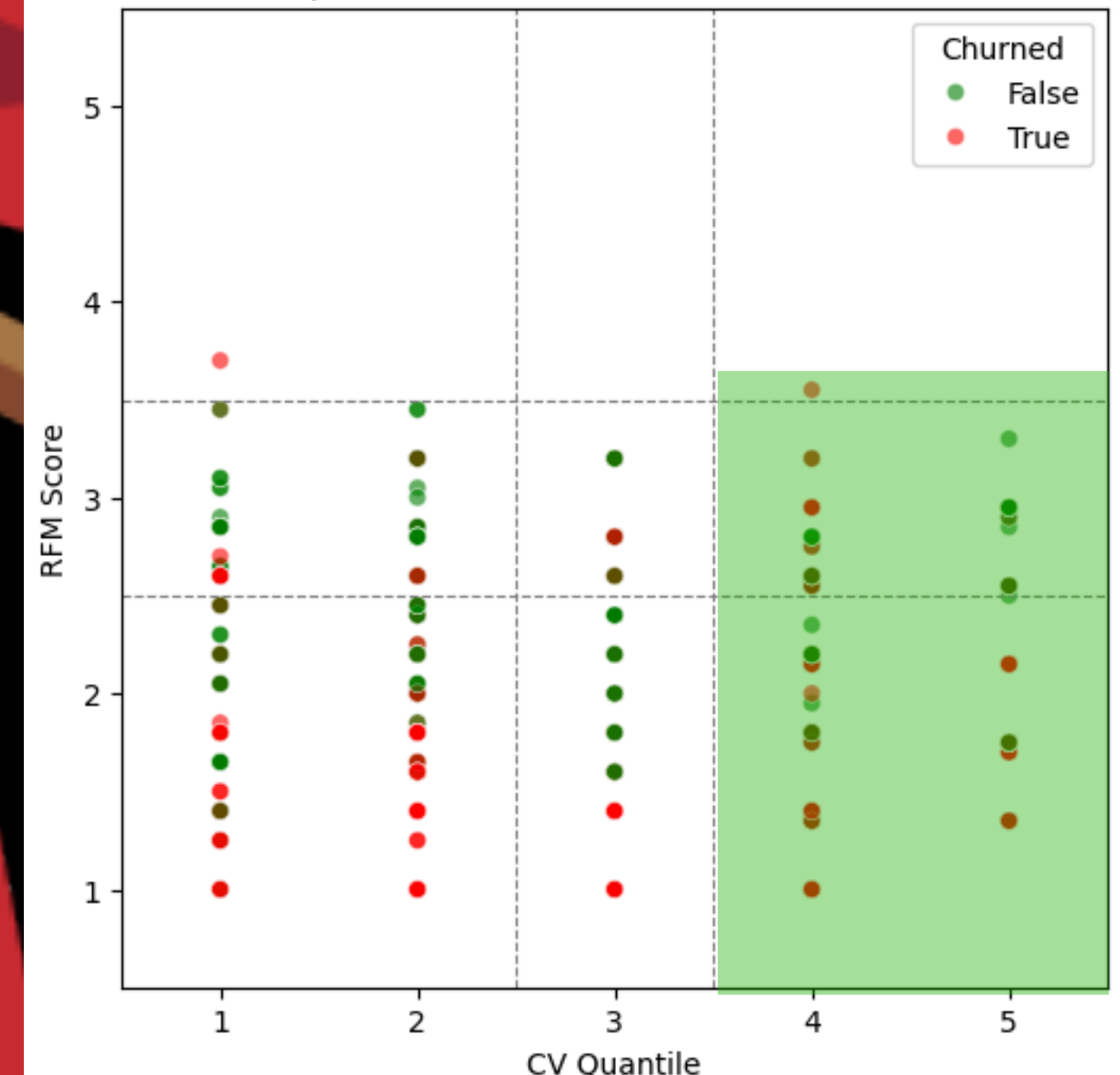
Bundle Savers Lite

110 Users

Growth Potential

- Goal: Reactivate the churned, improve frequency.
- Flash deals for hotel and flight bundles with attractive discounts

RFM x CV comparison for users classified as Bundle Savers Lite

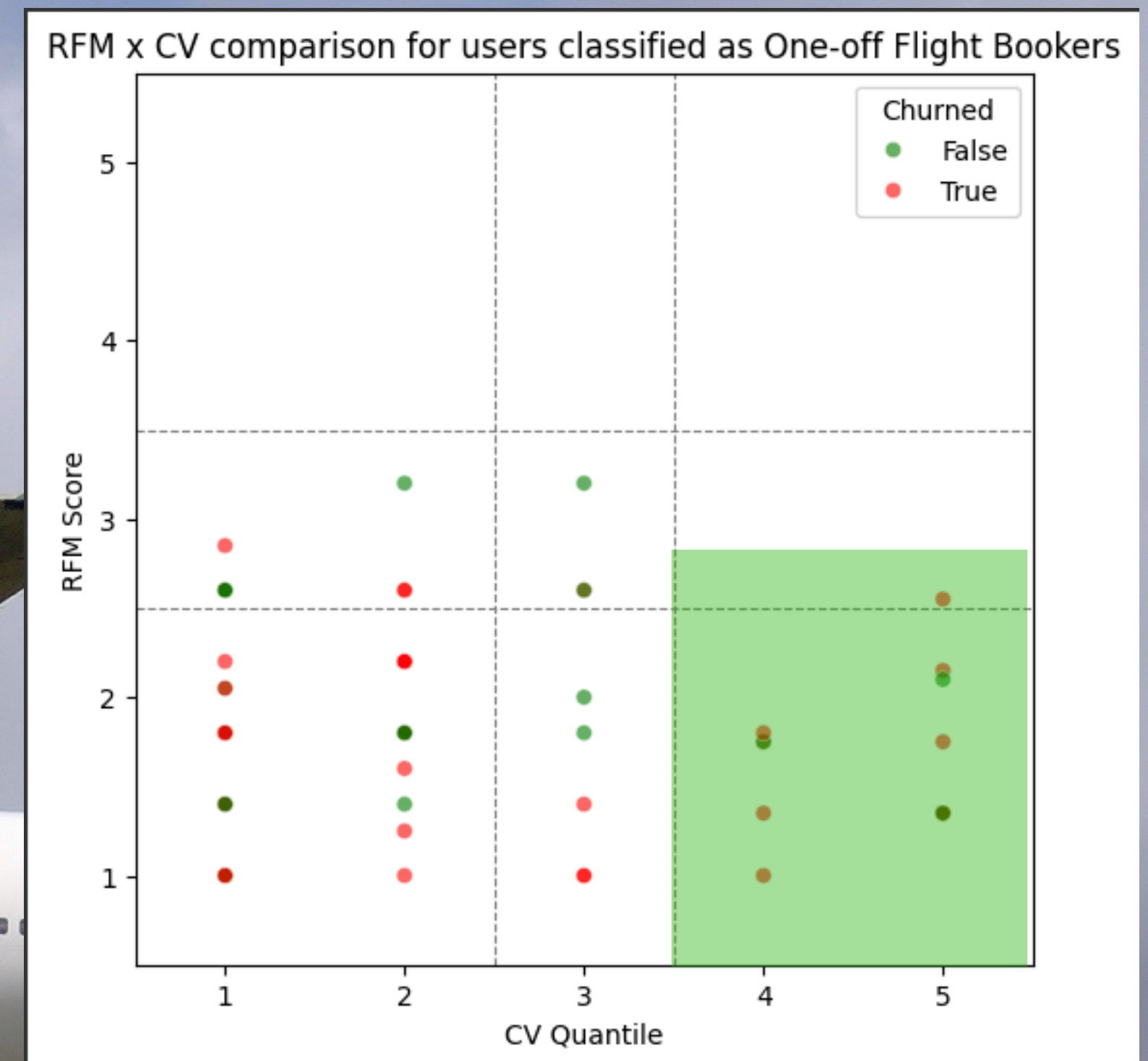




Almost all have booked only one trip, and only booked a flight
Above average booking lead time / planning horizon
Has some discount sensitivity
Have been away from the platform the longest among the clusters

One-off Flyers

They flew but once



One-off Flyers

64 Users

Growth Potential

- Goal: Reactivate the churned, improve frequency.
- Flash deals / limited time vouchers for airplane seat upgrades (e.g. voucher for leg room upgrade, economy to business class), free cancellation to lean into early booking behavior

Personas found by the Machine

Weak / Low Priority

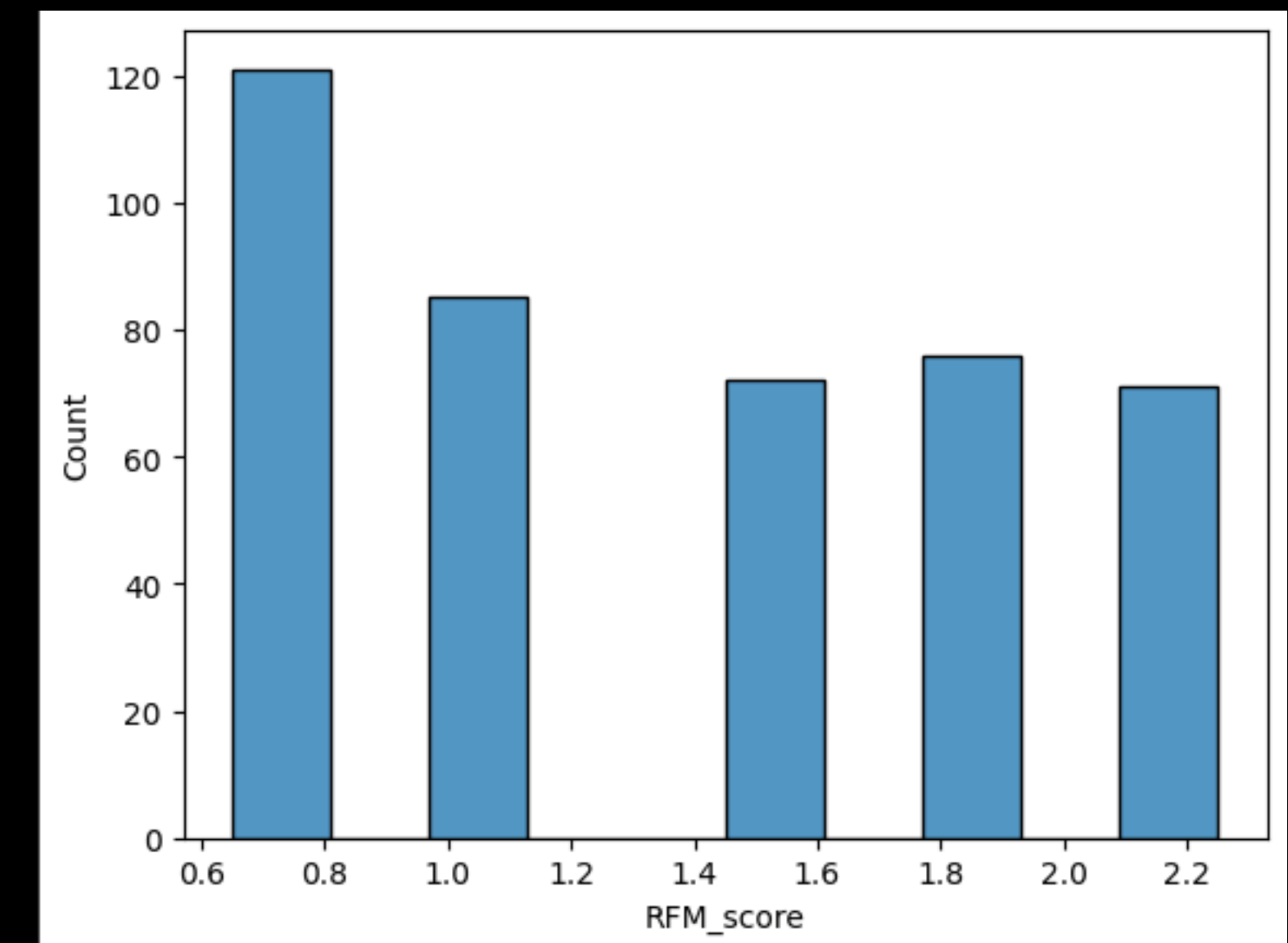


Almost all have booked only one trip, and only booked a hotel
 50% of the group booked trips of 6 days or less
 Has some discount sensitivity
 Overall low RFM and CV

One-off Hotel Bookers

190 Users

Never booked trips, like the Uninitiated
 Did not seem engaged during their sessions
 (lower session duration, lower clicks)
 Have been inactive for a longer time



The Uninterested

425 Users

Summary

Summary

- Each persona is further subdivided.
- A strategy is tailored to target users in each subdivision based on the persona's drivers.
- VIPs(High CV - High RFM) - try to maintain their position
- Core (Mid-High CV, Mid-High RFM) - try to push towards more frequent or higher tier spending
- Low (Low CV, Low RFM) - low-cost activation measures are more appropriate

Next Steps / Potential for Improvement

Next Steps / Potential for Improvement

- Run targeted marketing campaigns based on segmentation
- Determine most-effective perk for intersections of segments
- Try Unsupervised ML on the same limited parameters

Bonus

RFM Analysis

- Days since last session - recency
 - Weighted 0.4. Recency might signal likelihood of booking again soon
- Total spend - monetary
 - weighted 0.35. Big spenders are important but they may likely be the "whales" or VIP users, not the bulk of the user base.
- Total trips - frequency
 - weighted 0.25 because not booking often does not necessarily mean they are low-value

Customer Value (CV)

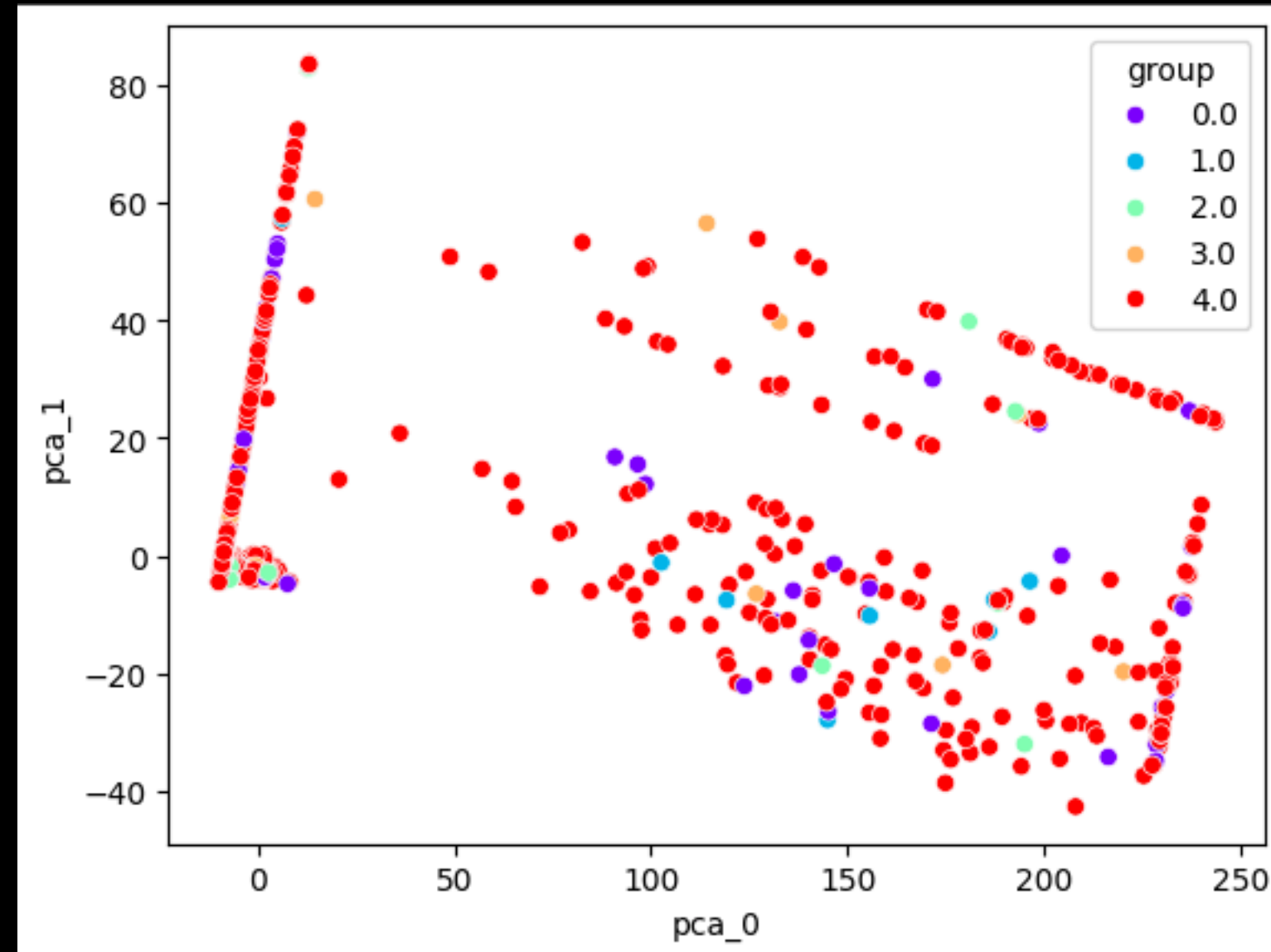
- Calculated per user
- $\text{Average Spend Per Trip} / \text{Average Number of Trips across all users}$

Churn Flagging

- Users whose last booking is twice or more as long as the average inter-purchase duration
- Average inter-purchase duration: ~45 days

ML Clustering

Lots of Correlations



- 95% variance between users captured within three Principal Components

Cluster Intersections

Traditional Segmentation

